

Tables

TABLE I: SAMPLE LUNG CANCER DATASET – PREPRUNING

GENDER	AGE	ANXIETY	ALCOHOL CONSUMING	LUNG_CANCER
M	69	2	2	YES

TABLE II: SAMPLE LUNG CANCER DATASET – POST PRUNING

GENDER	AGE	ANXIETY	ALCOHOL CONSUMING	LUNG_CANCER
1	69	1	1	1

TABLE III: AGE VARIABLE

Age Variable	
Mean	62.91
Median	62.50
Mode	64
Standard Deviation	8.38
Variance	70.21
Minimum Age	21
Maximum Age	87
Quartiles	Q1=57.75, Q2=62.50, Q3=69.00
Interquartile Range	11.25

TABLE IV: PRIMARY CATEGORICAL VARIABLES

S/N	Variable	Unique Values	Value	Frequency	Percentage (%)
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1	SMOKING	2	1	150	54.35
			0	126	45.65
2	ANXIETY	2	0	139	50.36
			1	137	49.64
3	PEER_PRESSURE	2	1	140	50.72
			0	136	49.28
4	ALCOHOL_CONSUMING	2	1	152	55.07
			0	124	44.93
5	LUNG_CANCER	2	1	238	86.23
			0	38	13.77

TABLE V: ALL CATEGORICAL VARIABLES IN DATASET

S/ N	Variable	Unique Values	Value	Frequency	Percentage (%)
1	GENDER	2	1	142	51.45
			0	134	48.55
2	SMOKING	2	1	150	54.35
			0	126	45.65
3	YELLOW_FINGERS	2	1	159	57.61
			0	117	42.39
4	ANXIETY	2	0	139	50.36
			1	137	49.64

5	PEER_PRESSURE	2	1	140	50.72
			0	136	49.28
6	CHRONIC_DISEASE	2	1	144	52.17
			0	132	47.83
7	FATIGUE	2	1	183	66.30
			0	93	33.70
8	ALLERGY	2	1	151	54.71
			0	125	45.29
9	WHEEZING	2	1	151	54.71
			0	125	45.29
10	ALCOHOL_CONSUMING	2	1	152	55.07
			0	124	44.93
11	COUGHING	2	1	159	57.61
			0	117	42.39
12	SHORTNESS_OF_BREATH	2	1	174	63.04
			0	102	36.96
13	SWALLOWING_DIFFICULTY	2	0	147	53.26
			1	129	46.74
14	CHEST_PAIN	2	1	154	55.80
			0	122	44.20
15	LUNG_CANCER	2	1	238	86.23

			0	38	13.77
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TABLE VI: TOP CORRELATIONS

Variable A	Variable B	Correlation
ANXIETY	YELLOW_FINGERS	0.5583
SWALLOWING_DIFFICULTY	ANXIETY	0.4788
ALCOHOL_CONSUMING	GENDER	0.4343
SHORTNESS_OF_BREATH	FATIGUE	0.4070
ALCOHOL_CONSUMING	ALLERGY	0.3781
CHEST_PAIN	GENDER	0.3615
COUGHING	WHEEZING	0.3537
LUNG_CANCER	ALLERGY	0.3336
SWALLOWING_DIFFICULTY	YELLOW_FINGERS	0.3333
SWALLOWING_DIFFICULTY	PEER_PRESSURE	0.3278

TABLE VII: MODELS EVALUATION WITH WAIC

Model	Elpd_waic
Base	-92.70
Full	-91.46
Interaction	-91.66

Figures:

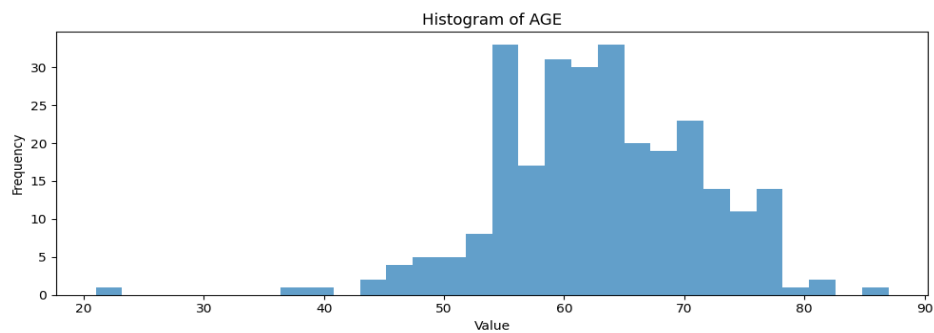


Fig. 1: Histogram of Age.

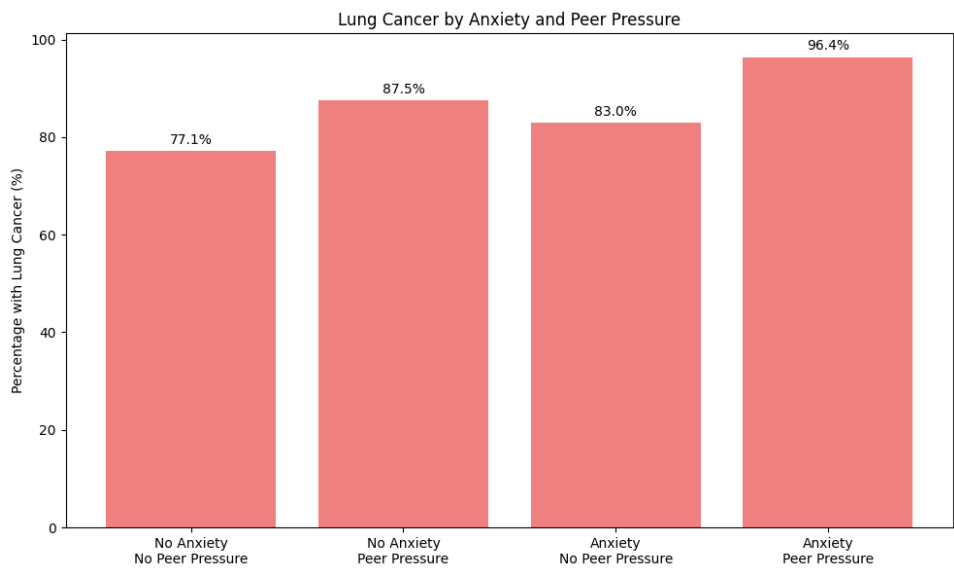


Fig. 2: Lung Cancer by Psychosocial Factors.

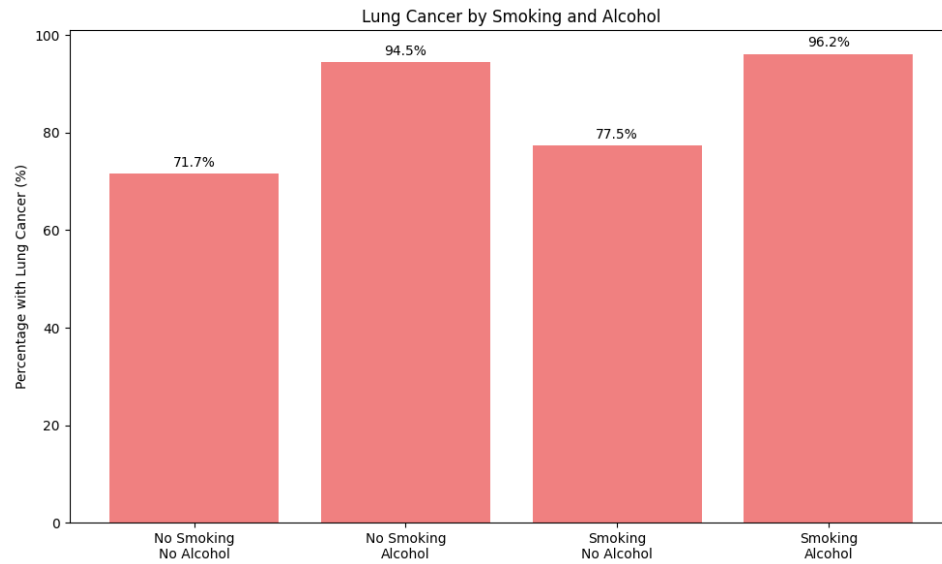


Fig. 3: Lung Cancer by Behavioural Factors.

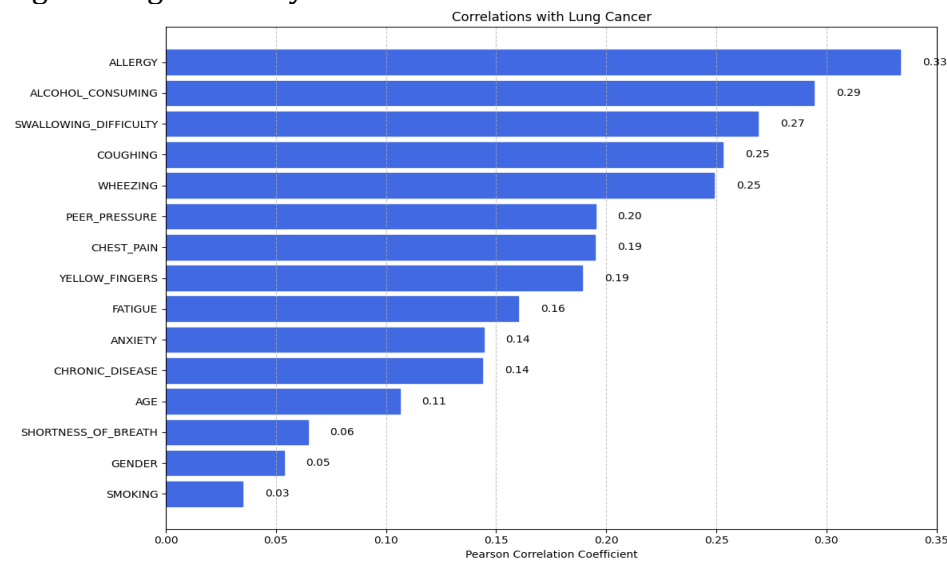


Fig. 4: Pearson's Correlations on All Variables Correlation with Lung Cancer.

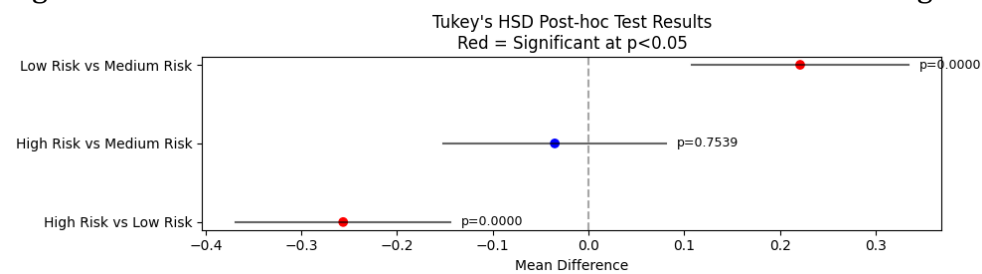


Fig. 5: Tukey's HSD Post-hoc Test Results on Significant Factors

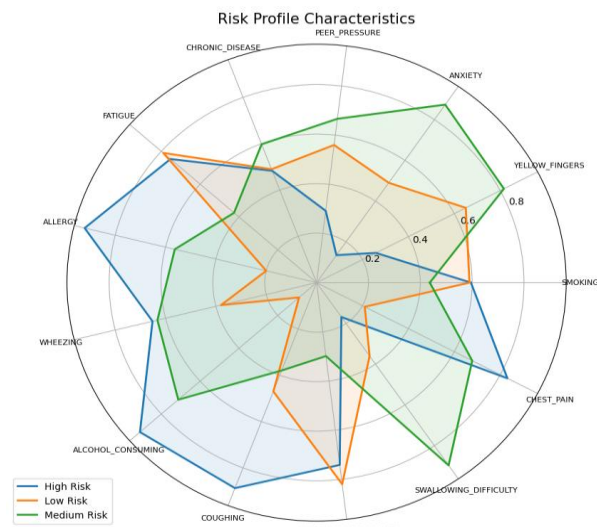


Fig. 6: Risk Profile Characteristics on All Variables using Clutter Chart

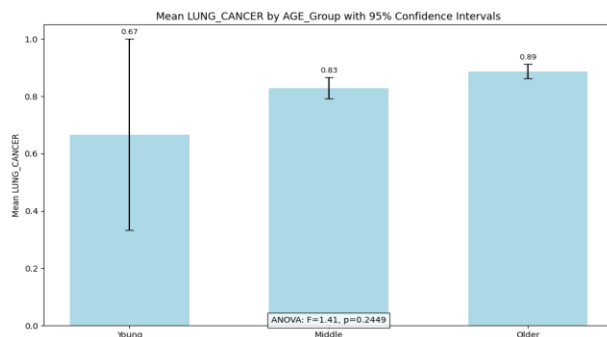


Fig. 7: Inferential Statistics Using Anova Test Anova Test Mean Lung Cancer by Age with 95% CI

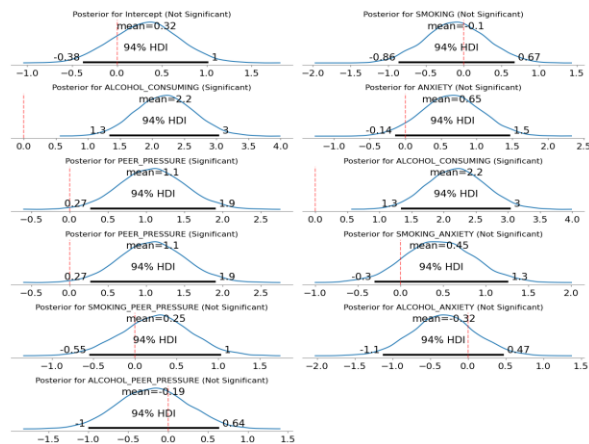


Fig. 8: Variables Posteriors and Significance

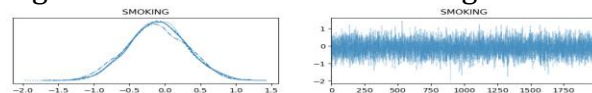


Fig. 9: MCMC Trace Diagnostics for SMOKING variable

Python Notebook Code: ([Repository Link](#))

This notebook is to carry out analysis on lung cancer and it's correlation based on the features shared by the dataset. The first step would be to visualise the data

```
import research as research
import matplotlib.pyplot as plt
```

```
data = research.load_and_preprocess_data('lung_cancer_survey.csv')
```

```
stats_results = research.descriptive_statistics(data)
```

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	\
0	M	69	1	2	2	1	
1	M	74	2	1	1	1	
2	F	59	1	1	1	2	
3	M	63	2	2	2	1	
4	F	63	1	2	1	1	

	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING
0	1	2	1	2	2	2
1	2	2	2	1	1	1
2	1	2	1	2	1	2
3	1	1	1	1	2	1
4	1	1	1	2	1	2

	SHORTNESS_OF_BREATH	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER
0	2	2	2	YES
1	2	2	2	YES
2	2	1	2	NO
3	1	2	2	NO
4	2	1	1	NO

Original shape: (309, 16)

After dropping duplicates: (276, 16)

Preprocessed data:

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	\
0	1	69	0	1	1	0	
1	1	74	1	0	0	0	
2	0	59	0	0	0	1	
3	1	63	1	1	1	0	
4	0	63	0	1	0	0	

	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING
0	0	1	0	1	1	1
1	1	1	1	0	0	0
2	0	1	0	1	0	1

3	0	0	0	0	1	0
4	0	0	0	1	0	1

	SHORTNESS_OF_BREATH	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER
0	1		1	1
1	1		1	1
2	1		1	0
3	0		1	0
4	1		0	0

Basic statistics:

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY \
count	276.000000	276.000000	276.000000	276.000000	276.000000
mean	0.514493	62.909420	0.543478	0.576087	0.496377
std	0.500698	8.379355	0.499011	0.495075	0.500895
min	0.000000	21.000000	0.000000	0.000000	0.000000
25%	0.000000	57.750000	0.000000	0.000000	0.000000
50%	1.000000	62.500000	1.000000	1.000000	0.000000
75%	1.000000	69.000000	1.000000	1.000000	1.000000
max	1.000000	87.000000	1.000000	1.000000	1.000000

	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING \
count	276.000000	276.000000	276.000000	276.000000	276.000000
mean	0.507246	0.521739	0.663043	0.547101	0.547101
std	0.500856	0.500435	0.473529	0.498681	0.498681
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	1.000000	1.000000	1.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000

	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH \
count	276.000000	276.000000	276.000000
mean	0.550725	0.576087	0.630435
std	0.498324	0.495075	0.483564
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	1.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000

	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER
count	276.000000	276.000000	276.000000
mean	0.467391	0.557971	0.862319
std	0.499842	0.497530	0.345191
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	1.000000
50%	0.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000

max	1.000000	1.000000	1.000000
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Columns:

```
Index(['GENDER', 'AGE', 'SMOKING', 'YELLOW_FINGERS', 'ANXIETY',  
      'PEER_PRESSURE', 'CHRONIC_DISEASE', 'FATIGUE', 'ALLERGY', 'WHEEZING',  
      'ALCOHOL_CONSUMING', 'COUGHING', 'SHORTNESS_OF_BREATH',  
      'SWALLOWING_DIFFICULTY', 'CHEST_PAIN', 'LUNG_CANCER'],  
      dtype='object')
```

Descriptive statistics

```
print("Generating Visualisations")
```

```
print("\nHistograms for Continuous Variables")
```

```
hist_fig = stats_results['visualisations']['histograms']()  
plt.show()
```

```
print("\nBoxplots for Continuous Variables")
```

```
box_fig = stats_results['visualisations']['boxplots']()  
plt.show()
```

```
print("\nBar Charts for Categorical Variables")
```

```
bar_fig = stats_results['visualisations']['bar_charts']()  
plt.show()
```

```
print("\nCorrelation Heatmap")
```

```
corr_fig = stats_results['visualisations']['correlation_heatmap']()  
plt.show()
```

```
print("\nVisualisation of Lung Cancer Correlations")
```

```
lung_corr_fig = stats_results['visualisations']['lung_cancer_correlations']()  
plt.show()
```

```
print("\nKey Factor Correlation Heatmap")
```

```
factor_corr_fig = stats_results['visualisations']['factor_correlations']()  
plt.show()
```

```
print("\nDetailed Results for Report")
```

```
print("\nContinuous Variable Statistics")
```

```
for var, stats in stats_results['continuous'].items():  
    print(f"\n {var}")  
    print(f"- Mean: {stats['mean']:.2f}")  
    print(f"- Median: {stats['median']:.2f}")  
    print(f"- Mode: {stats['mode']}")  
    print(f"- Standard Deviation: {stats['std_dev']:.2f}")  
    print(f"- Variance: {stats['variance']:.2f}")  
    print(f"- Minimum: {stats['min']}")
```

```

    print(f"- Maximum: {stats['max']}")
    print(f"- Quartiles: Q1={stats['quartiles'][0]:.2f},
Q2={stats['quartiles'][1]:.2f}, Q3={stats['quartiles'][2]:.2f}")
    print(f"- Interquartile Range (IQR): {stats['IQR']:.2f}")

print("\n Categorical Variable Statistics")
for var, stats in stats_results['categorical'].items():
    print(f"\n {var}")
    print(f"- Unique Values: {stats['unique_values']}")
    print(f"- Mode: {stats['mode']['value']} (Count:
{stats['mode']['count']}, {stats['mode']['percentage']:.2f}%)")

    print(f"- Frequency Table:")
    freq_table = stats['frequency_table']
    print("| Value | Frequency | Percentage |")
    print("|-----|-----|-----|")
    for idx, row in freq_table.iterrows():
        print(f"| {idx} | {int(row['frequency'])} | {row['percentage']:.2f}%
|")

print("\nCorrelation Analysis")
print("\n Top Correlations")
print("| Variable 1 | Variable 2 | Correlation |")
print("|-----|-----|-----|")
for var1, var2, corr in stats_results['correlation']['top_correlations']:
    print(f"| {var1} | {var2} | {corr:.4f} |")

print("\n Strongest Correlations")
abs_corr_pairs = [(v1, v2, corr) for v1, v2, corr in
stats_results['correlation']['top_correlations']]
abs_corr_pairs.sort(key=lambda x: abs(x[2]), reverse=True)

print("\nStrongest Positive Correlations:")
print("| Variable 1 | Variable 2 | Correlation |")
print("|-----|-----|-----|")
for var1, var2, corr in [p for p in abs_corr_pairs if p[2] > 0][:5]:
    print(f"| {var1} | {var2} | {corr:.4f} |")

print("\nStrongest Negative Correlations:")
print("| Variable 1 | Variable 2 | Correlation |")
print("|-----|-----|-----|")
for var1, var2, corr in [p for p in abs_corr_pairs if p[2] < 0][:5]:
    print(f"| {var1} | {var2} | {corr:.4f} |")

print("\nCorrelations with Lung Cancer")
print("| Variable | Correlation with Lung Cancer |")
print("|-----|-----|")
for var, corr in

```

```

stats_results['correlation']['top_lung_cancer_correlations']:
    print(f"| {var} | {corr:.4f} |")

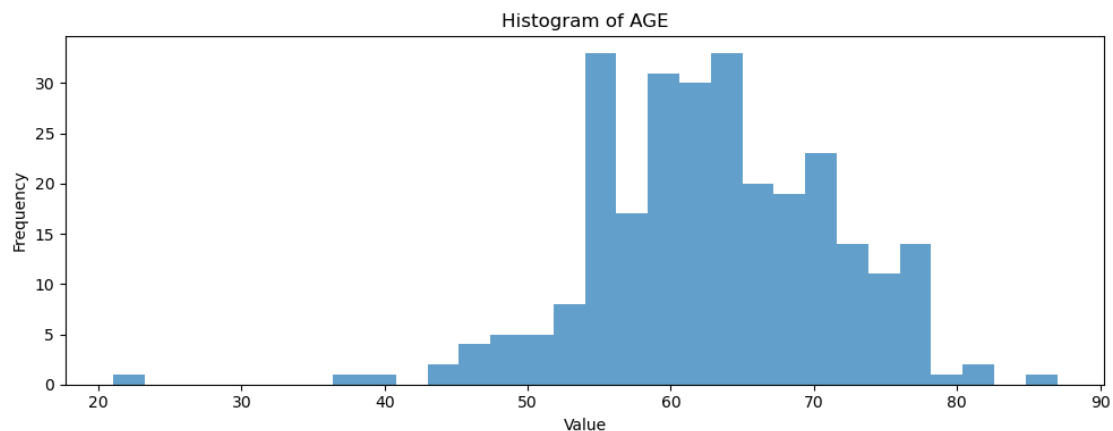
print("\n Psychosocial Factors and Lung Cancer")
print("| Factor | Correlation with Lung Cancer |")
print("|-----|-----|")
for factor, corr in
stats_results['correlation']['psychosocial_correlations'].items():
    print(f"| {factor} | {corr:.4f} |")

print("\n Behavioral Habits and Lung Cancer")
print("| Habit | Correlation with Lung Cancer |")
print("|-----|-----|")
for habit, corr in
stats_results['correlation']['behavioral_correlations'].items():
    print(f"| {habit} | {corr:.4f} |")

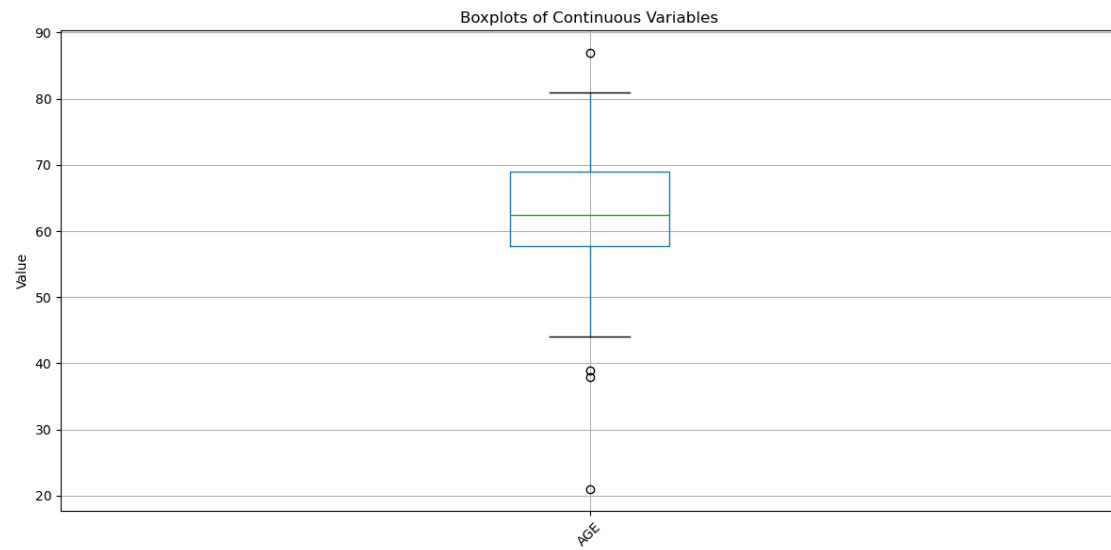
```

Generating Visualisations

Histograms for Continuous Variables

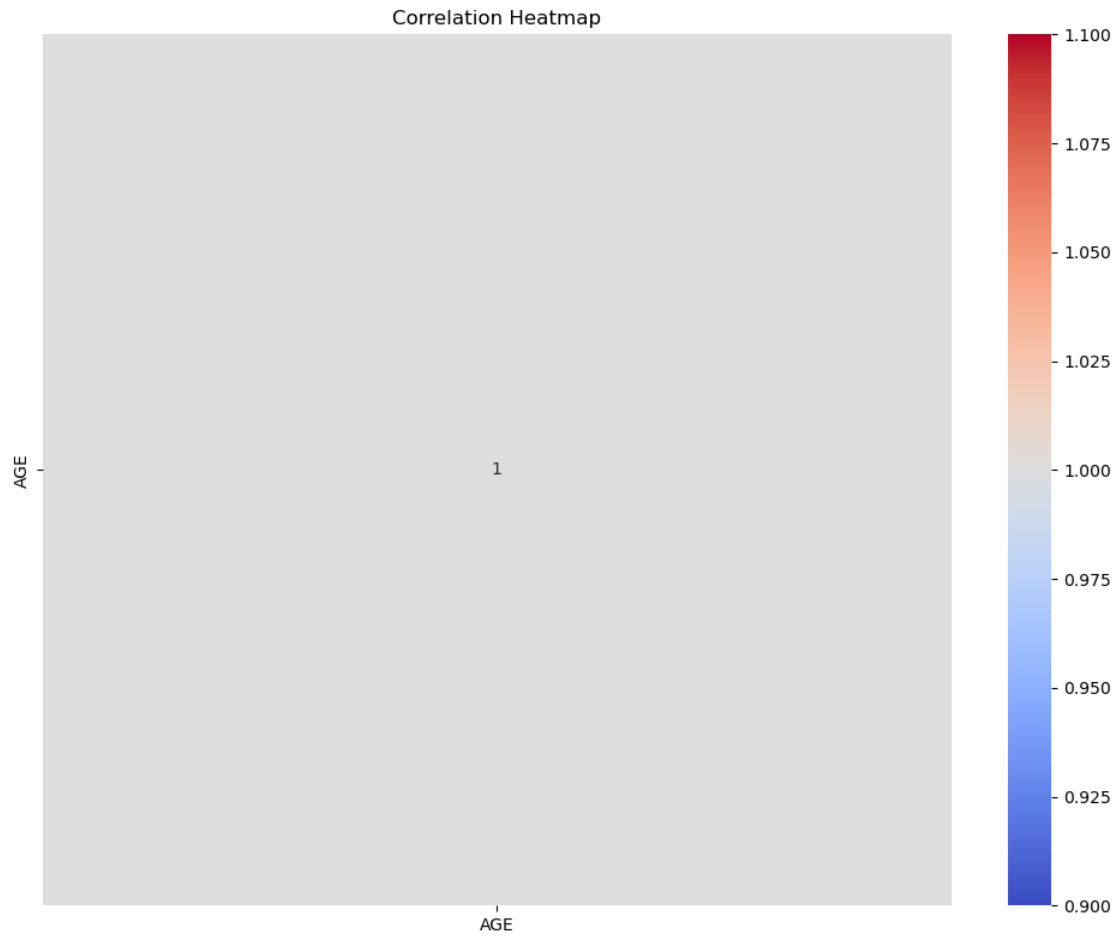


Boxplots for Continuous Variables

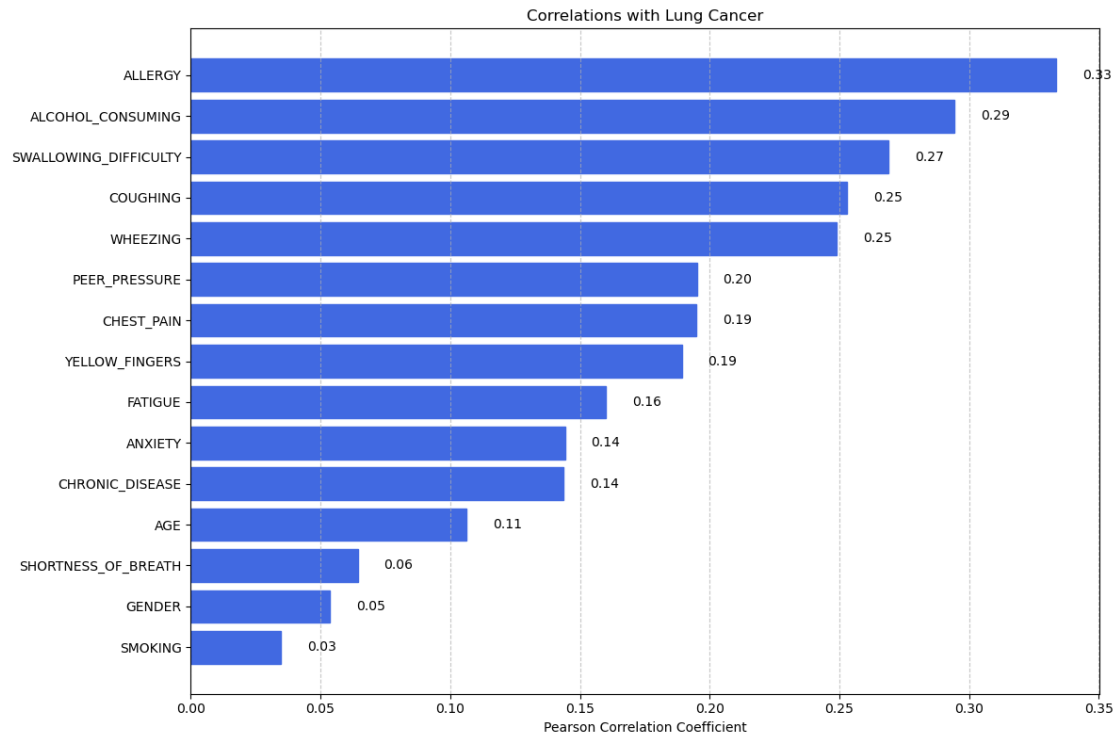


Bar Charts for Categorical Variables

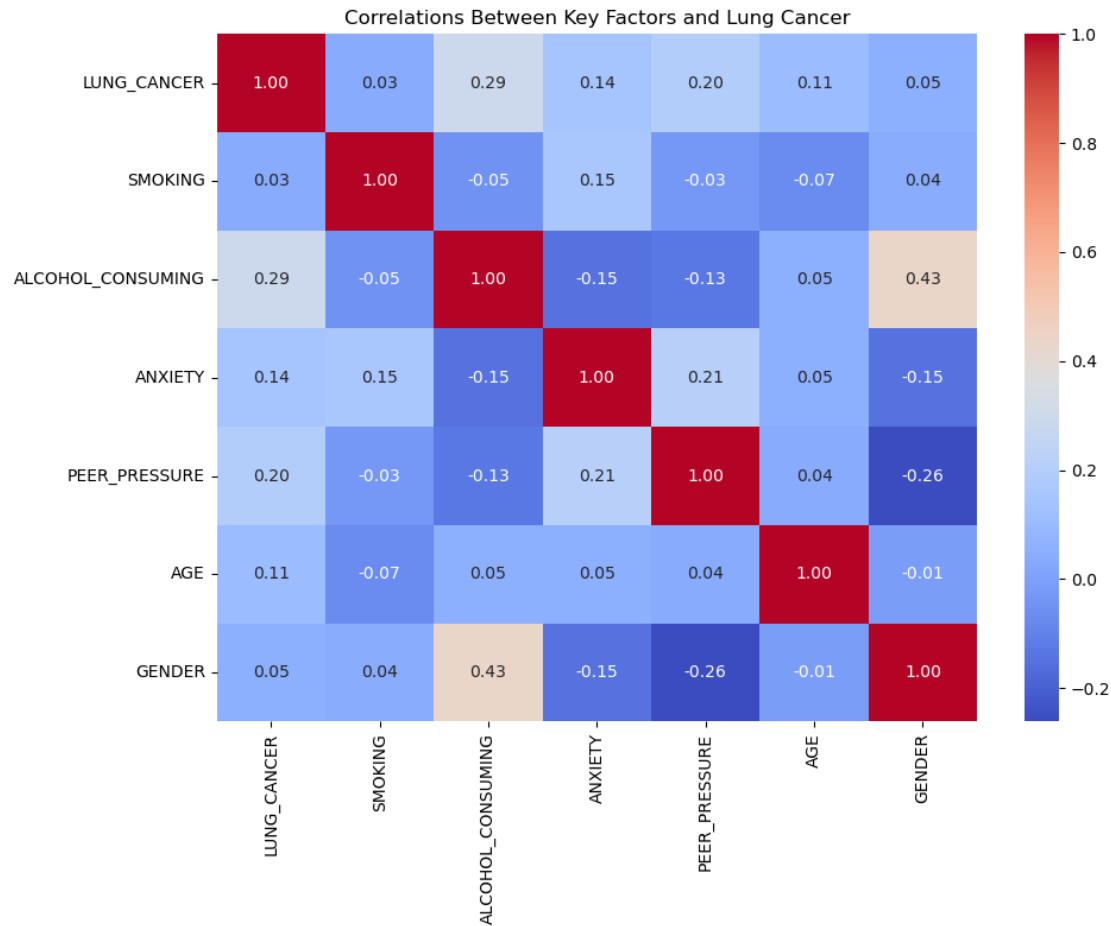
Correlation Heatmap



Visualisation of Lung Cancer Correlations



Key Factor Correlation Heatmap



Detailed Results for Report

Continuous Variable Statistics

AGE

- Mean: 62.91
- Median: 62.50
- Mode: 64
- Standard Deviation: 8.38
- Variance: 70.21
- Minimum: 21
- Maximum: 87
- Quartiles: Q1=57.75, Q2=62.50, Q3=69.00
- Interquartile Range (IQR): 11.25

Categorical Variable Statistics

GENDER

- Unique Values: 2
- Mode: 1 (Count: 142, 51.45%)

- Frequency Table:

Value	Frequency	Percentage
1	142	51.45%
0	134	48.55%

SMOKING

- Unique Values: 2

- Mode: 1 (Count: 150, 54.35%)

- Frequency Table:

Value	Frequency	Percentage
1	150	54.35%
0	126	45.65%

YELLOW_FINGERS

- Unique Values: 2

- Mode: 1 (Count: 159, 57.61%)

- Frequency Table:

Value	Frequency	Percentage
1	159	57.61%
0	117	42.39%

ANXIETY

- Unique Values: 2

- Mode: 0 (Count: 139, 50.36%)

- Frequency Table:

Value	Frequency	Percentage
0	139	50.36%
1	137	49.64%

PEER_PRESSURE

- Unique Values: 2

- Mode: 1 (Count: 140, 50.72%)

- Frequency Table:

Value	Frequency	Percentage
1	140	50.72%
0	136	49.28%

CHRONIC_DISEASE

- Unique Values: 2

- Mode: 1 (Count: 144, 52.17%)

- Frequency Table:

Value	Frequency	Percentage
1	144	52.17%

| 0 | 132 | 47.83% |

FATIGUE

- Unique Values: 2
- Mode: 1 (Count: 183, 66.30%)

- Frequency Table:

Value	Frequency	Percentage
1	183	66.30%
0	93	33.70%

ALLERGY

- Unique Values: 2
- Mode: 1 (Count: 151, 54.71%)

- Frequency Table:

Value	Frequency	Percentage
1	151	54.71%
0	125	45.29%

WHEEZING

- Unique Values: 2
- Mode: 1 (Count: 151, 54.71%)

- Frequency Table:

Value	Frequency	Percentage
1	151	54.71%
0	125	45.29%

ALCOHOL_CONSUMING

- Unique Values: 2
- Mode: 1 (Count: 152, 55.07%)

- Frequency Table:

Value	Frequency	Percentage
1	152	55.07%
0	124	44.93%

COUGHING

- Unique Values: 2
- Mode: 1 (Count: 159, 57.61%)

- Frequency Table:

Value	Frequency	Percentage
1	159	57.61%
0	117	42.39%

SHORTNESS_OF_BREATH

- Unique Values: 2

- Mode: 1 (Count: 174, 63.04%)

- Frequency Table:

Value	Frequency	Percentage
1	174	63.04%
0	102	36.96%

SWALLOWING_DIFFICULTY

- Unique Values: 2

- Mode: 0 (Count: 147, 53.26%)

- Frequency Table:

Value	Frequency	Percentage
0	147	53.26%
1	129	46.74%

CHEST_PAIN

- Unique Values: 2

- Mode: 1 (Count: 154, 55.80%)

- Frequency Table:

Value	Frequency	Percentage
1	154	55.80%
0	122	44.20%

LUNG_CANCER

- Unique Values: 2

- Mode: 1 (Count: 238, 86.23%)

- Frequency Table:

Value	Frequency	Percentage
1	238	86.23%
0	38	13.77%

Correlation Analysis

Top Correlations

Variable 1	Variable 2	Correlation
ANXIETY	YELLOW_FINGERS	0.5583
SWALLOWING_DIFFICULTY	ANXIETY	0.4788
ALCOHOL_CONSUMING	GENDER	0.4343
SHORTNESS_OF_BREATH	FATIGUE	0.4070
ALCOHOL_CONSUMING	ALLERGY	0.3781
CHEST_PAIN	GENDER	0.3615
COUGHING	WHEEZING	0.3537
LUNG_CANCER	ALLERGY	0.3336
SWALLOWING_DIFFICULTY	YELLOW_FINGERS	0.3333
SWALLOWING_DIFFICULTY	PEER_PRESSURE	0.3278

Strongest Correlations

Strongest Positive Correlations:

Variable 1	Variable 2	Correlation
ANXIETY	YELLOW_FINGERS	0.5583
SWALLOWING_DIFFICULTY	ANXIETY	0.4788
ALCOHOL_CONSUMING	GENDER	0.4343
SHORTNESS_OF_BREATH	FATIGUE	0.4070
ALCOHOL_CONSUMING	ALLERGY	0.3781

Strongest Negative Correlations:

Variable 1	Variable 2	Correlation

Correlations with Lung Cancer

Variable	Correlation with Lung Cancer
ALLERGY	0.3336
ALCOHOL_CONSUMING	0.2944
SWALLOWING_DIFFICULTY	0.2689
COUGHING	0.2530
WHEEZING	0.2491
PEER_PRESSURE	0.1951
CHEST_PAIN	0.1949
YELLOW_FINGERS	0.1892
FATIGUE	0.1601
ANXIETY	0.1443

Psychosocial Factors and Lung Cancer

Factor	Correlation with Lung Cancer
ANXIETY	0.1443
PEER_PRESSURE	0.1951

Behavioral Habits and Lung Cancer

Habit	Correlation with Lung Cancer
SMOKING	0.0349
ALCOHOL_CONSUMING	0.2944

Cancer Features Distribution Visualisations

```
print("\n Cancer Distribution Visualisations")
```

```
results = research.lung_cancer_features(data)
data = results[0]
cancer_by_feature = results[1]
```

```

psychosocial_dist = results[2]
behavioral_dist = results[3]
print("\n Overall Lung Cancer Distribution")
cancer_dist_fig = research.plot_cancer_distribution(data)
plt.show()

print("\n Cancer Distribution by Individual Factors")

for feature in ['SMOKING', 'ALCOHOL_CONSUMING', 'ANXIETY', 'PEER_PRESSURE']:
    feature_fig = research.plot_cancer_by_feature(data, feature)
    plt.show()

print("\n Cancer Distribution by Combined Factors")
psycho_fig = research.plot_combined_factors(data, psychosocial_dist,
'psychosocial')
plt.show()

behav_fig = research.plot_combined_factors(data, behavioral_dist,
'behavioral')
plt.show()

```

Cancer Distribution Visualisations

Cancer Distribution:

Lung Cancer (Yes): 86.23%

No Lung Cancer (No): 13.77%

Cancer Distribution by Features:

SMOKING:

No: 84.92% have lung cancer

Yes: 87.33% have lung cancer

YELLOW_FINGERS:

No: 78.63% have lung cancer

Yes: 91.82% have lung cancer

ANXIETY:

No: 81.29% have lung cancer

Yes: 91.24% have lung cancer

PEER_PRESSURE:

No: 79.41% have lung cancer

Yes: 92.86% have lung cancer

CHRONIC_DISEASE:

No: 81.06% have lung cancer

Yes: 90.97% have lung cancer

FATIGUE:

No: 78.49% have lung cancer
Yes: 90.16% have lung cancer

ALLERGY:

No: 73.60% have lung cancer
Yes: 96.69% have lung cancer

WHEEZING:

No: 76.80% have lung cancer
Yes: 94.04% have lung cancer

ALCOHOL_CONSUMING:

No: 75.00% have lung cancer
Yes: 95.39% have lung cancer

COUGHING:

No: 76.07% have lung cancer
Yes: 93.71% have lung cancer

SHORTNESS_OF_BREATH:

No: 83.33% have lung cancer
Yes: 87.93% have lung cancer

SWALLOWING_DIFFICULTY:

No: 77.55% have lung cancer
Yes: 96.12% have lung cancer

CHEST_PAIN:

No: 78.69% have lung cancer
Yes: 92.21% have lung cancer

Complex Cancer Distribution - Psychosocial and Behavioral Factors:

Psychosocial Factors and Lung Cancer:

Anxiety_PeerPressure | Count | % of Total | % with Lung Cancer

No_No	83.0	30.07%	77.11%
No_Yes	56.0	20.29%	87.50%
Yes_No	53.0	19.20%	83.02%
Yes_Yes	84.0	30.43%	96.43%

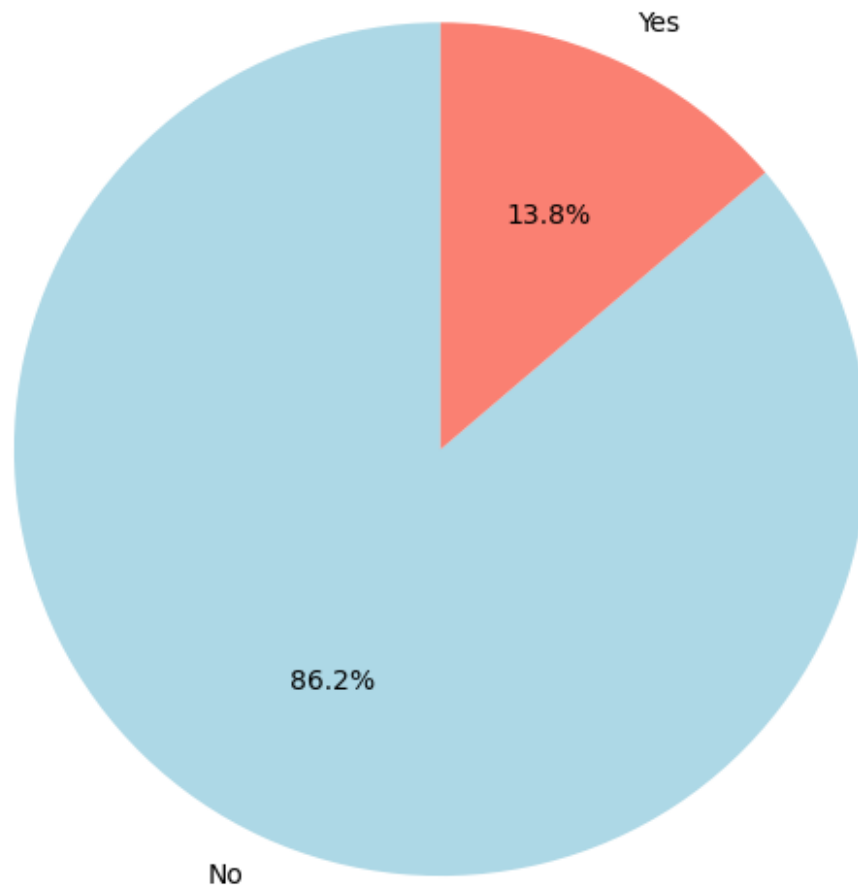
Behavioral Habits and Lung Cancer:

Smoking_Alcohol | Count | % of Total | % with Lung Cancer

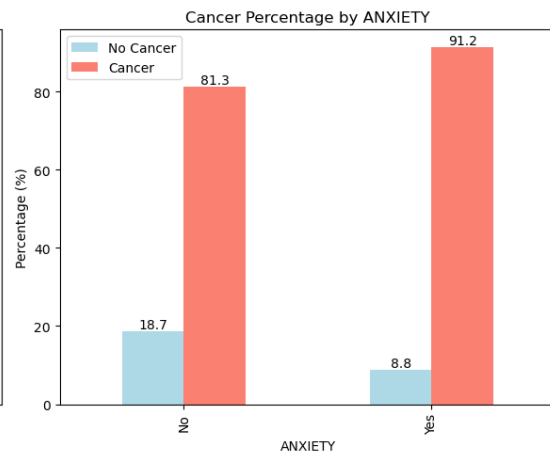
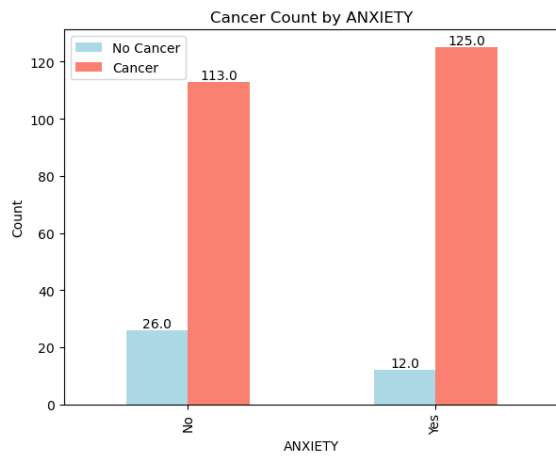
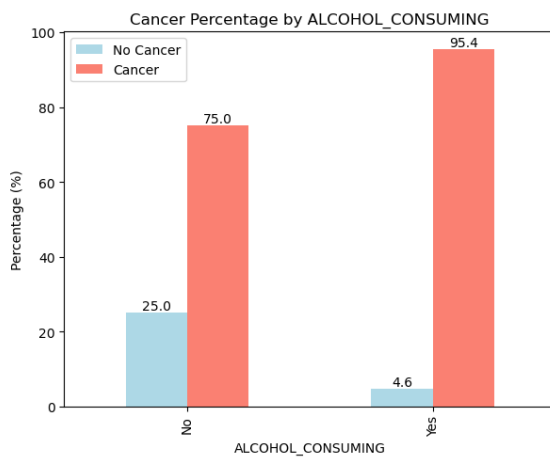
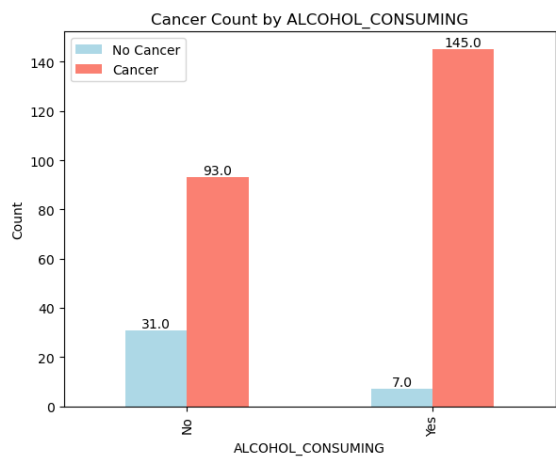
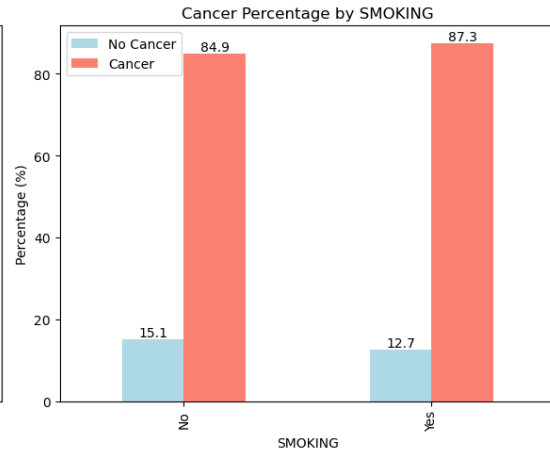
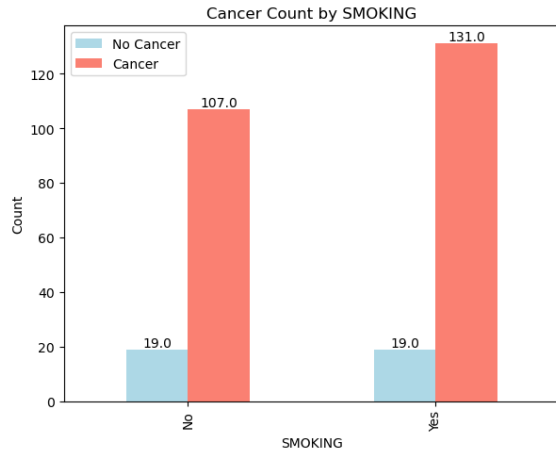
No_No		53.0		19.20%		71.70%
No_Yes		73.0		26.45%		94.52%
Yes_No		71.0		25.72%		77.46%
Yes_Yes		79.0		28.62%		96.20%

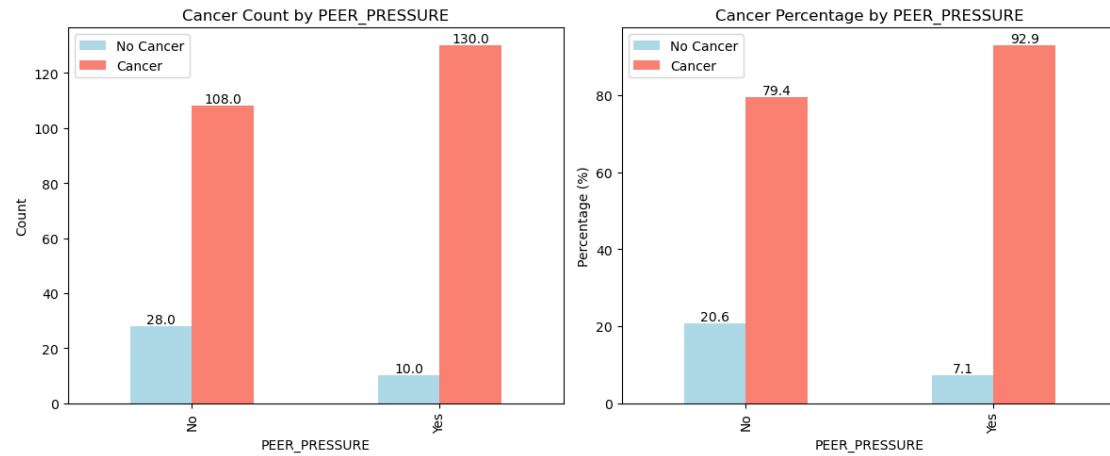
Overall Lung Cancer Distribution

Lung Cancer Distribution

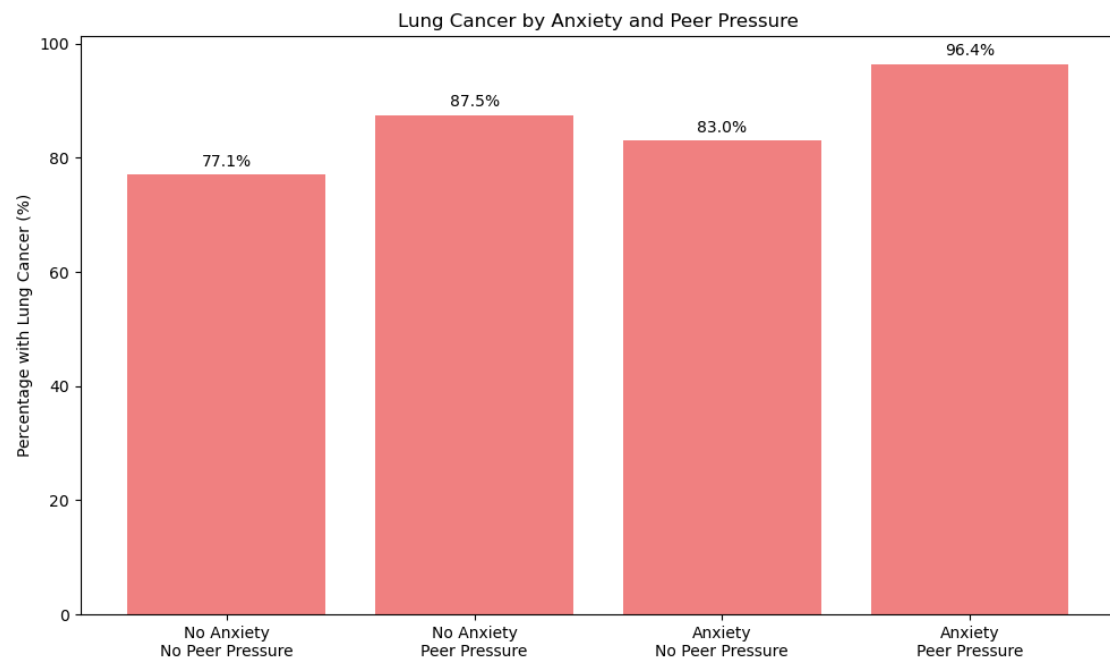


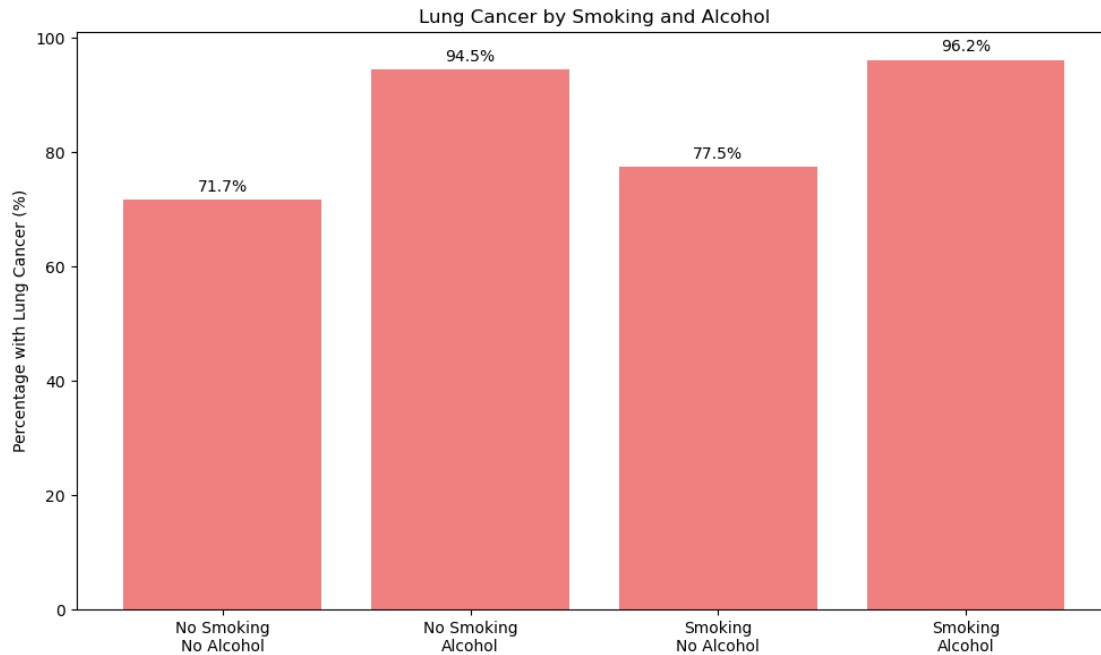
Cancer Distribution by Individual Factors





Cancer Distribution by Combined Factors





Inferential Statistics Analysis

Logistic Regression with Interaction terms

```
print("\n Logistic Regression with Interaction Terms")
from inferential_statistics import inferential_statistics as infer_stats
model, model_data, interaction_results, viz_functions =
infer_stats.logistic_regression_analysis(data)
```

```
print("\n Coefficient Plot")
coef_fig = viz_functions['coefficients']()
plt.show()
```

```
print("\n ROC Curve")
roc_fig = viz_functions['roc_curve']()
plt.show()
```

```
print("\n Confusion Matrix")
cm_fig = viz_functions['confusion_matrix']()
plt.show()
```

```
print("\n Interaction Effects")
int_fig = viz_functions['interaction_effects']()
plt.show()
```

```
print("\n Significant Interactions")
print("| Variable | Coefficient | Odds Ratio | p-value |")
print("|-----|-----|-----|-----|")
for _, row in interaction_results[interaction_results['p-value'] <
```

```
0.05].iterrows():
    print(f"| {row['Variable']} | {row['Coefficient']:.4f} | {row['Odds
Ratio']:.4f} | {row['p-value']:.4f} |")
```

Logistic Regression with Interaction Terms

Logistic Regression Analysis with Interaction Terms

Creating Interaction Terms

Fitting Logistic Regression Models

Likelihood Ratio Tests

Base vs Psychosocial Interactions: Chi2=13.24, df=4.0, p-value=0.0102
Psychosocial vs Full Model: Chi2=11.05, df=8.0, p-value=0.1991

Selected Model: Model with psychosocial interactions

Model Summary

Log-Likelihood: -79.17
AIC: 180.35
BIC: 220.17
Pseudo R-squared: 0.2842

Key Coefficients and Wald Tests

	coef	std err	z	P> z	[0.025
0.975]					

Intercept	0.2136	0.429	0.498	0.618	-0.627
1.054					
AGE_C	0.0143	0.024	0.602	0.547	-0.032
0.061					
GENDER	-0.5395	0.474	-1.139	0.255	-1.468
0.389					
SMOKING	-0.9047	0.649	-1.394	0.163	-2.177
0.368					
ALCOHOL_CONSUMING	4.1713	0.828	5.035	0.000	2.548
5.795					
ANXIETY	0.6486	0.662	0.979	0.327	-0.649
1.947					
PEER_PRESSURE	1.4326	0.647	2.215	0.027	0.165
2.700					
SMOKING_ANXIETY	1.6402	0.882	1.860	0.063	-0.088
3.368					
SMOKING_PEER_PRESSURE	0.5187	0.889	0.584	0.560	-1.223

2.261					
ALCOHOL_ANXIETY	-2.2142	0.978	-2.264	0.024	-4.131
-0.298					
ALCOHOL_PEER_PRESSURE	-1.7913	0.970	-1.846	0.065	-3.693
0.110					

=====

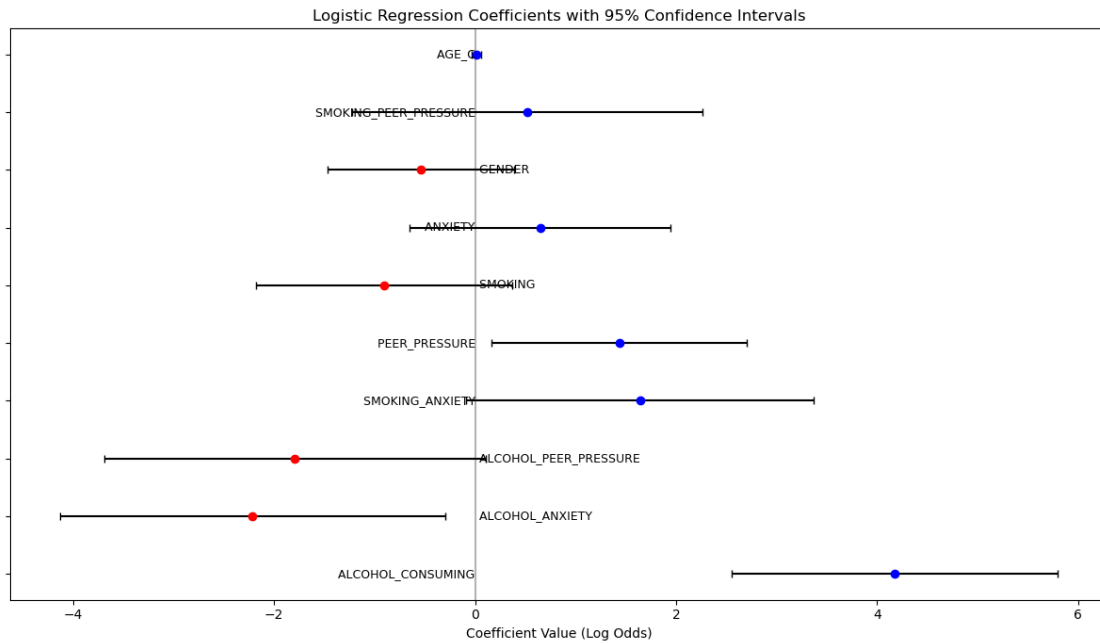
=====

Classification Report					
	precision	recall	f1-score	support	
0	0.50	0.21	0.30	38	
1	0.88	0.97	0.92	238	
accuracy				0.86	276
macro avg				0.69	276
weighted avg				0.83	276

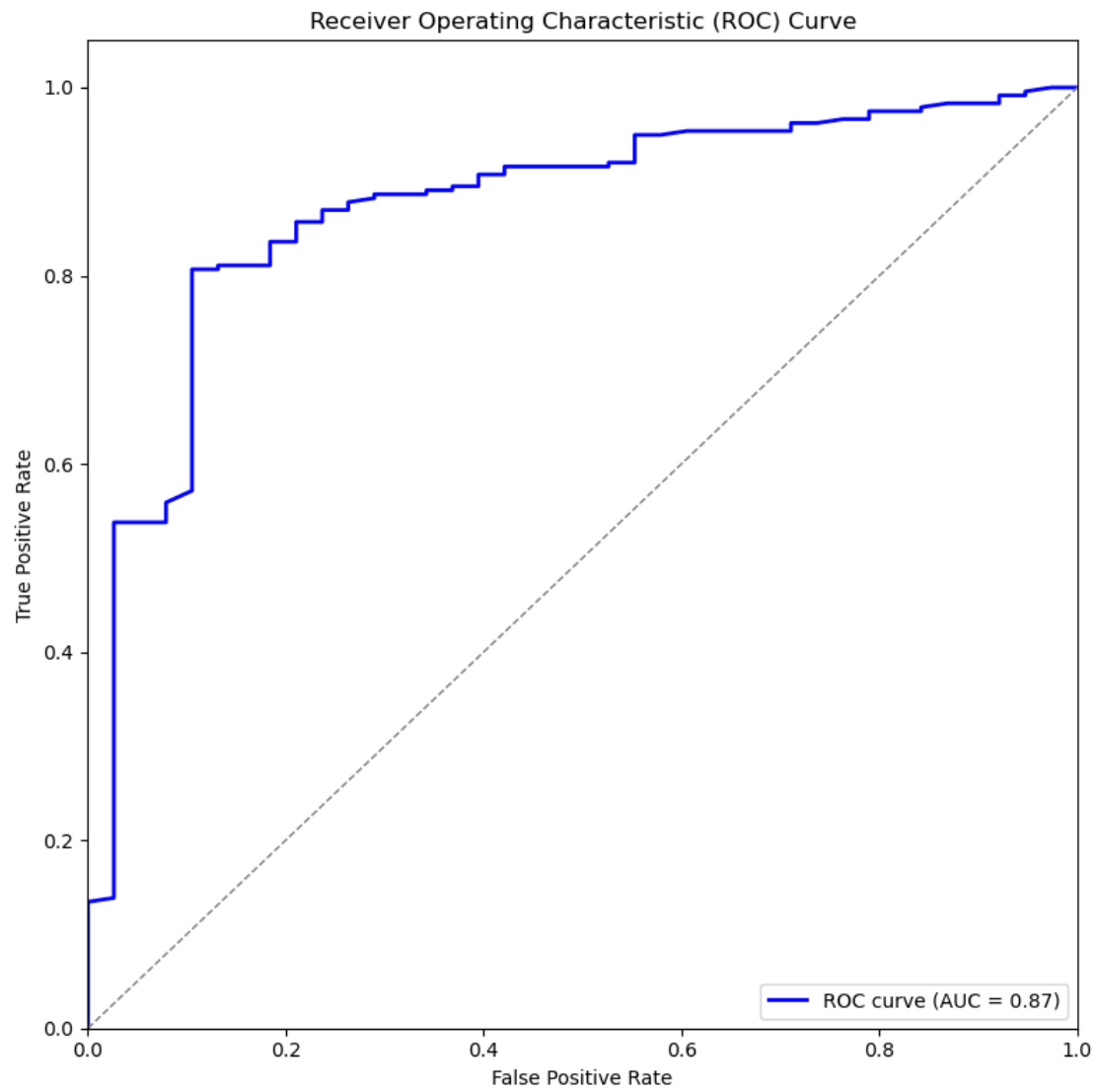
Significant Interactions					
	Variable	Coefficient	Odds Ratio	Std Error	z-value \
0	ALCOHOL_CONSUMING	4.171294	64.799232	0.828401	5.035354
4	ALCOHOL_ANXIETY	-2.214208	0.109240	0.977816	-2.264443
1	PEER_PRESSURE	1.432562	4.189419	0.646866	2.214621
p-value					
0		4.769656e-07			
4		2.354689e-02			
1		2.678609e-02			

Coefficient Plot

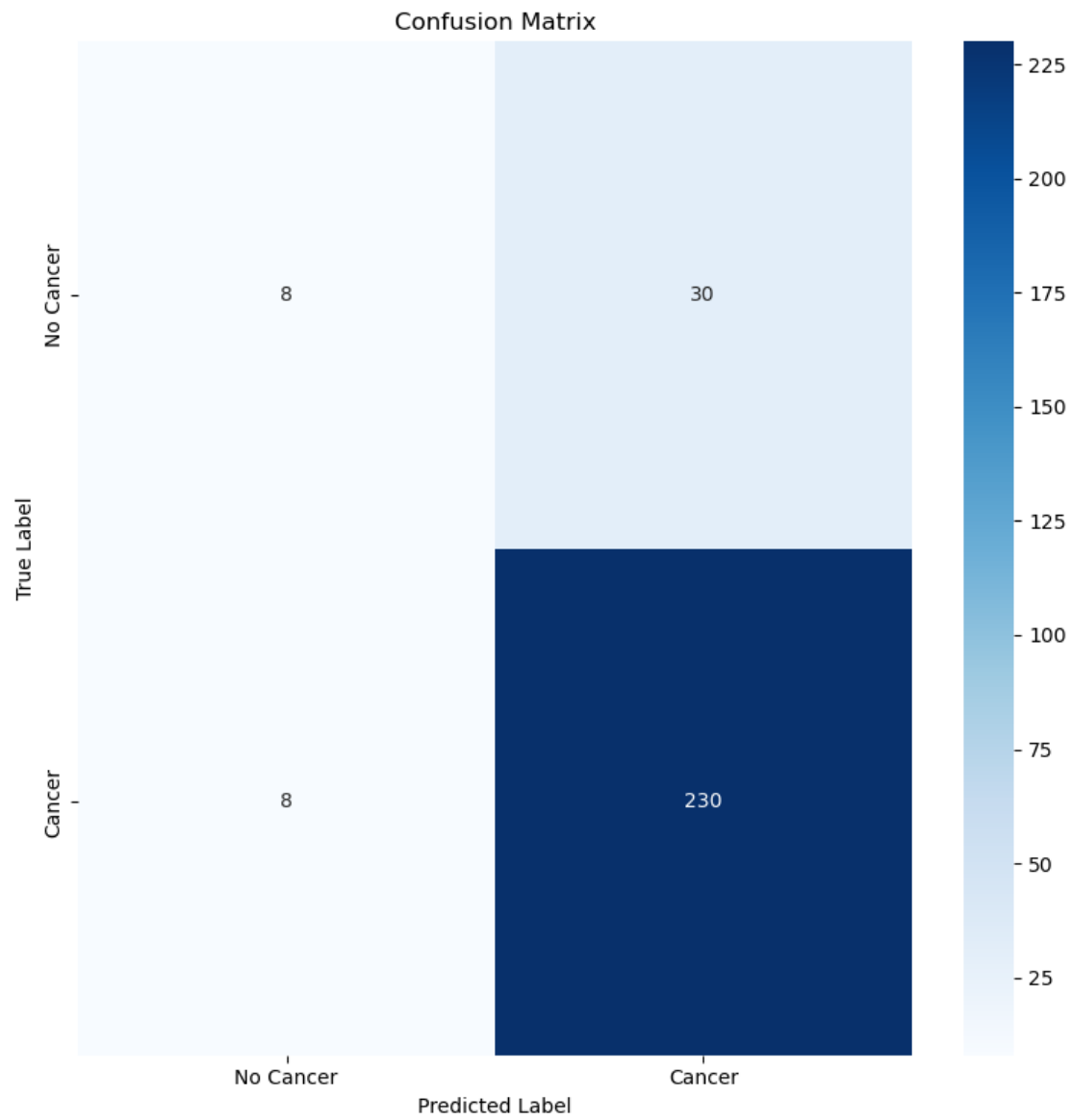
c:\ProgramData\anaconda3\Lib\site-packages\statsmodels\base\model.py:607:
 ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check
 mle_retvals
 warnings.warn("Maximum Likelihood optimization failed to "



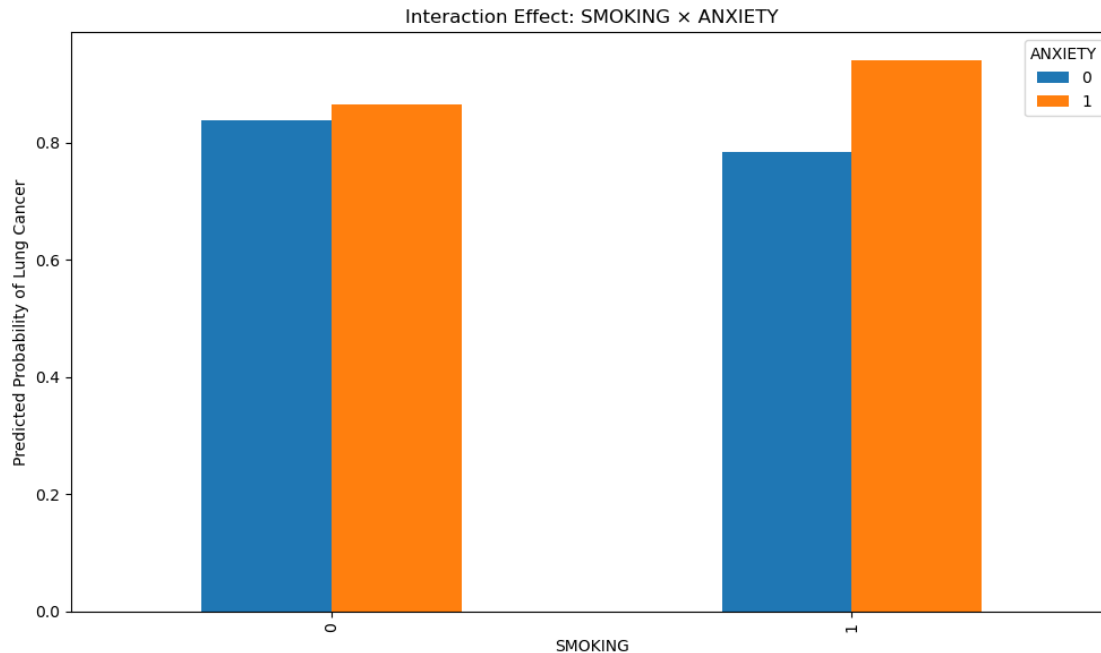
ROC Curve



Confusion Matrix



Interaction Effects



Significant Interactions

Variable	Coefficient	Odds Ratio	p-value
ALCOHOL_CONSUMING	4.1713	64.7992	0.0000
ALCOHOL_ANXIETY	-2.2142	0.1092	0.0235
PEER_PRESSURE	1.4326	4.1894	0.0268

Confidence Interval

```
print("\n Coefficient Estimates with 95% Confidence Intervals")
ci_table = viz_functions['coefficient_intervals']()

print("\n Main Effects")
print("| Variable | Coefficient | 95% CI | Odds Ratio | OR 95% CI | p-value |")
print("|-----|-----|-----|-----|-----|-----|")
for idx, row in ci_table[ci_table['Type'] == 'Main Effect'].iterrows():
    if idx == 'Intercept':
        continue
    print(f"| {idx} | {row['Coefficient']:.4f} | {row['95% CI']} | {row['Odds Ratio']:.4f} | {row['OR 95% CI']} | {row['p-value']:.4f} |")

print("\n Interaction Terms")
print("| Variable | Coefficient | 95% CI | Odds Ratio | OR 95% CI | p-value |")
print("|-----|-----|-----|-----|-----|-----|")
for idx, row in ci_table[ci_table['Type'] == 'Interaction'].iterrows():
```



```

    if row['p-value'] < 0.05:
        print(f" | **{idx}** | **{row['Coefficient']:.4f}** | **{row['95% CI']:.4f}** | **{row['Odds Ratio']:.4f}** | **{row['OR 95% CI']:.4f}** | **{row['p-value']:.4f}** |")
    else:
        print(f" | {idx} | {row['Coefficient']:.4f} | {row['95% CI']:.4f} | {row['Odds Ratio']:.4f} | {row['OR 95% CI']:.4f} | {row['p-value']:.4f} |")

print("\n Interpretation of Confidence Intervals")
print("- Confidence intervals that don't include 0 for coefficients (or 1 for odds ratios) indicate statistically significant effects at  $\alpha=0.05$ ")
print("- Wider confidence intervals indicate less precision in the estimate")
print("- For interaction terms, significant CIs suggest the effect of one variable depends on the level of another")

```

Coefficient Estimates with 95% Confidence Intervals

Main Effects

Variable	Coefficient	95% CI	Odds Ratio	OR 95% CI	p-value
SMOKING	-0.9047	(-2.1771, 0.3677)	0.4047	(0.1134, 1.4444)	0.1635
GENDER	-0.5395	(-1.4682, 0.3892)	0.5830	(0.2303, 1.4757)	0.2549
ANXIETY	0.6486	(-0.6493, 1.9465)	1.9129	(0.5224, 7.0042)	0.3273
AGE_C	0.0143	(-0.0323, 0.0610)	1.0144	(0.9682, 1.0629)	0.5475

Interaction Terms

Variable	Coefficient	95% CI	Odds Ratio	OR 95% CI	p-value
ALCOHOL_CONSUMING	**4.1713**	**(2.5477, 5.7949)**	**64.7992**	**(12.7771, 328.6293)**	**0.0000**
ALCOHOL_ANXIETY	**2.2142**	**(-4.1307, -0.2977)**	**0.1092**	**(0.0161, 0.7425)**	**0.0235**
PEER_PRESSURE	**1.4326**	**(0.1647, 2.7004)**	**4.1894**	**(1.1791, 14.8856)**	**0.0268**
SMOKING_ANXIETY	1.6402	(-0.0881, 3.3684)	5.1562	(0.9157, 29.0334)	0.0629
ALCOHOL_PEER_PRESSURE	-1.7913	(-3.6930, 0.1104)	0.1667	(0.0249, 1.1168)	0.0649
SMOKING_PEER_PRESSURE	0.5187	(-1.2235, 2.2609)	1.6799	(0.2942, 9.5919)	0.5595

Interpretation of Confidence Intervals

- Confidence intervals that don't include 0 for coefficients (or 1 for odds ratios) indicate statistically significant effects at $\alpha=0.05$
- Wider confidence intervals indicate less precision in the estimate

- For interaction terms, significant CIs suggest the effect of one variable depends on the level of another

Anova test

```
print("\n Analysis of Variance (ANOVA)")
from inferential_statistics import inferential_statistics as infer_stats
anova_results, viz_functions = infer_stats.anova_analysis(data)
```

```
print("\n Box Plots by Group")
age_boxplot = viz_functions['group_boxplots']('age_groups')
plt.show()
```

```
gender_boxplot = viz_functions['group_boxplots']('gender')
plt.show()
```

```
smoking_alcohol_boxplot = viz_functions['group_boxplots']('smoking_alcohol')
plt.show()
```

```
print("\n Mean Comparisons")
age_means = viz_functions['group_means']('age_groups')
plt.show()
```

```
risk_means = viz_functions['group_means']('risk_profiles')
plt.show()
```

```
print("\n Post-hoc Test Results")
posthoc_fig = viz_functions['posthoc_results']('risk_profiles')
plt.show()
```

```
if 'risk_profiles' in viz_functions:
    print("\n Risk Profile Characteristics")
    profile_fig = viz_functions['risk_profiles']()
    plt.show()
```

```
significant_groups = []
for group_name, results in anova_results.items():
    if results.get('p_value', 1) < 0.05:
        significant_groups.append((group_name, results))
```

```
if significant_groups:
    print("\n Significant Group Differences Found")
    print("| Grouping Variable | F-value | p-value | Significant Pairwise")
    print("Comparisons |")
    print("-----|-----|-----|-----")
    print("-----|")
```

```

for group_name, results in significant_groups:
    pairwise_text = "None"
    if 'posthoc_result' in results:
        sig_pairs = []
        posthoc = results['posthoc_result']
        for i in range(len(posthoc.pvalues)):
            if posthoc.pvalues[i] < 0.05:
                row = posthoc._results_table.data[i+1]
                group1, group2 = row[0], row[1]
                sig_pairs.append(f"{group1} vs {group2}")

        if sig_pairs:
            pairwise_text = ", ".join(sig_pairs)

    print(f"| {results['group_var']} | {results['f_value']:.2f} |
{results['p_value']:.4f} | {pairwise_text} |")
else:
    print("\n No Significant Group Differences Found")

```

Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA)

Creating Age Groups

ANOVA: Lung Cancer by Age Group

Group statistics for AGE_Group:

	count	mean	std
AGE_Group			
Young	3	0.666667	0.577350
Middle	105	0.828571	0.378691
Older	168	0.886905	0.317656

ANOVA results: $F(2, 273) = 1.4140$, $p = 0.2449$

ANOVA is not significant. No post-hoc tests needed.

Creating Risk Profiles via Clustering

```

c:\Users\princ\source\repos\school\Reaearch
Methods\preassessment\Isaac\inferential_statistics.py:757: FutureWarning: The
default of observed=False is deprecated and will be changed to True in a
future version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
    group_stats = data.groupby(group_var)[dependent_var].agg(['count', 'mean',
'std'])
c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1429:

```

UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=2.
warnings.warn(

ANOVA: Lung Cancer by Risk Profile

Group statistics for Risk_Level:

	count	mean	std
Risk_Level			
High Risk	89	0.966292	0.181499
Low Risk	100	0.710000	0.456048
Medium Risk	87	0.931034	0.254864

ANOVA results: $F(2, 273) = 17.3378$, $p = 0.0000$

ANOVA is significant. Performing Tukey's HSD post-hoc test...

Tukey's HSD results:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
High Risk	Low Risk	-0.2563	0.0	-0.3684	-0.1442	True
High Risk	Medium Risk	-0.0353	0.7539	-0.1512	0.0807	False
Low Risk	Medium Risk	0.221	0.0	0.1083	0.3338	True

Significant pairwise comparisons ($p < 0.05$):

High Risk vs Low Risk: Mean diff = -0.2563, $p = 0.0000$

High Risk vs Medium Risk: Mean diff = -0.0353, $p = 0.7539$

Low Risk vs Medium Risk: Mean diff = 0.2210, $p = 0.0000$

ANOVA: Lung Cancer by Gender

Group statistics for GENDER:

	count	mean	std
GENDER			
0	134	0.843284	0.364897
1	142	0.880282	0.325781

ANOVA results: $F(1, 274) = 0.7914$, $p = 0.3745$

ANOVA is not significant. No post-hoc tests needed.

ANOVA: Lung Cancer by Smoking and Alcohol Status

Group statistics for Smoking_Alcohol:

	count	mean	std
Smoking_Alcohol			
Alcohol Only	73	0.945205	0.229154
Both	79	0.962025	0.192356
Neither	53	0.716981	0.454776
Smoking Only	71	0.774648	0.420788

ANOVA results: $F(3, 272) = 8.9771$, $p = 0.0000$

ANOVA is significant. Performing Tukey's HSD post-hoc test...

Tukey's HSD results:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
  group1      group2  meandiff p-adj  lower  upper  reject
-----
Alcohol Only      Both    0.0168 0.9894 -0.1221  0.1558  False
Alcohol Only      Neither -0.2282 0.0009 -0.3827 -0.0738  True
Alcohol Only Smoking Only -0.1706 0.0118 -0.3132 -0.0279  True
      Both      Neither   -0.245 0.0002 -0.397 -0.0931  True
      Both Smoking Only -0.1874 0.0035 -0.3273 -0.0474  True
      Neither Smoking Only  0.0577 0.7725 -0.0977  0.213  False
-----
```

Significant pairwise comparisons ($p < 0.05$):

Alcohol Only vs Both: Mean diff = 0.0168, $p = 0.9894$

Alcohol Only vs Neither: Mean diff = -0.2282, $p = 0.0009$

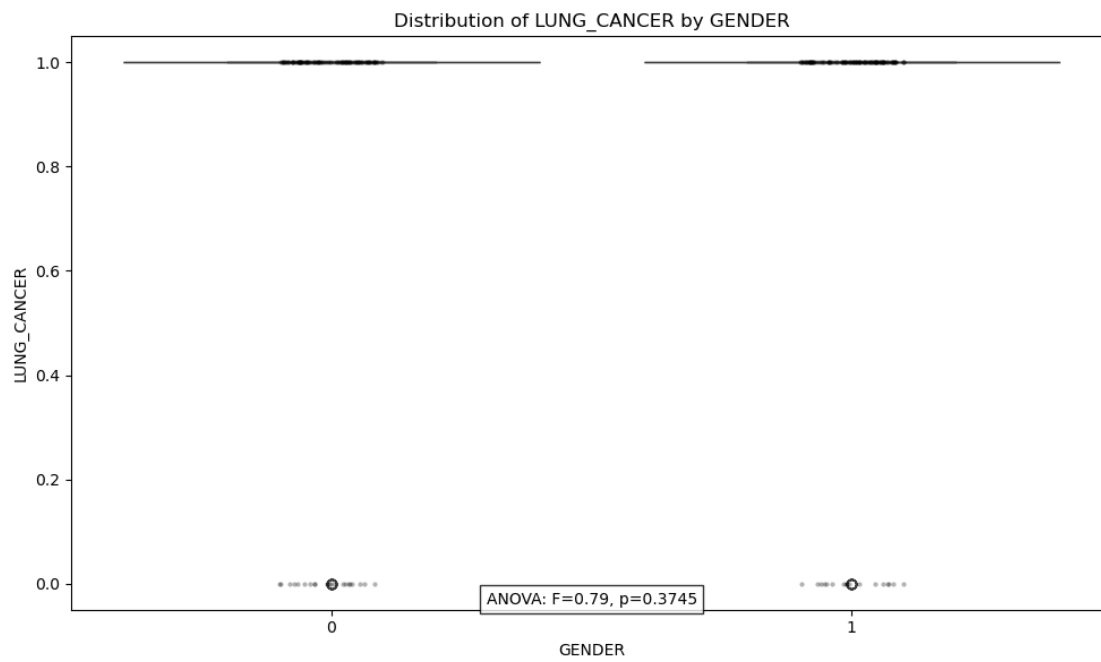
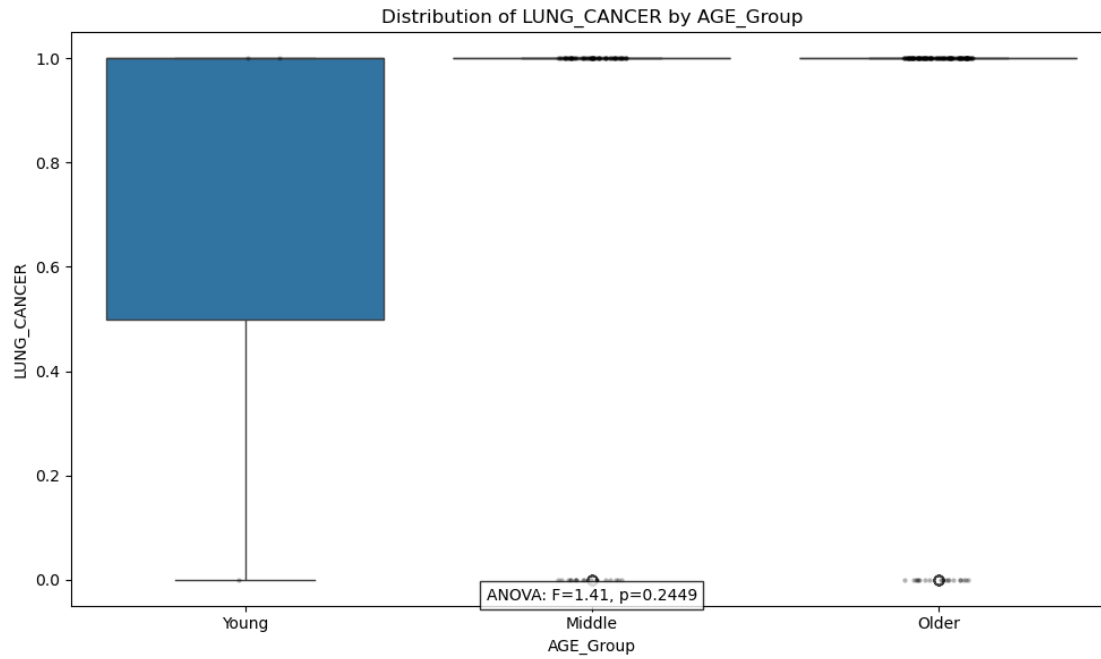
Alcohol Only vs Smoking Only: Mean diff = -0.1706, $p = 0.0118$

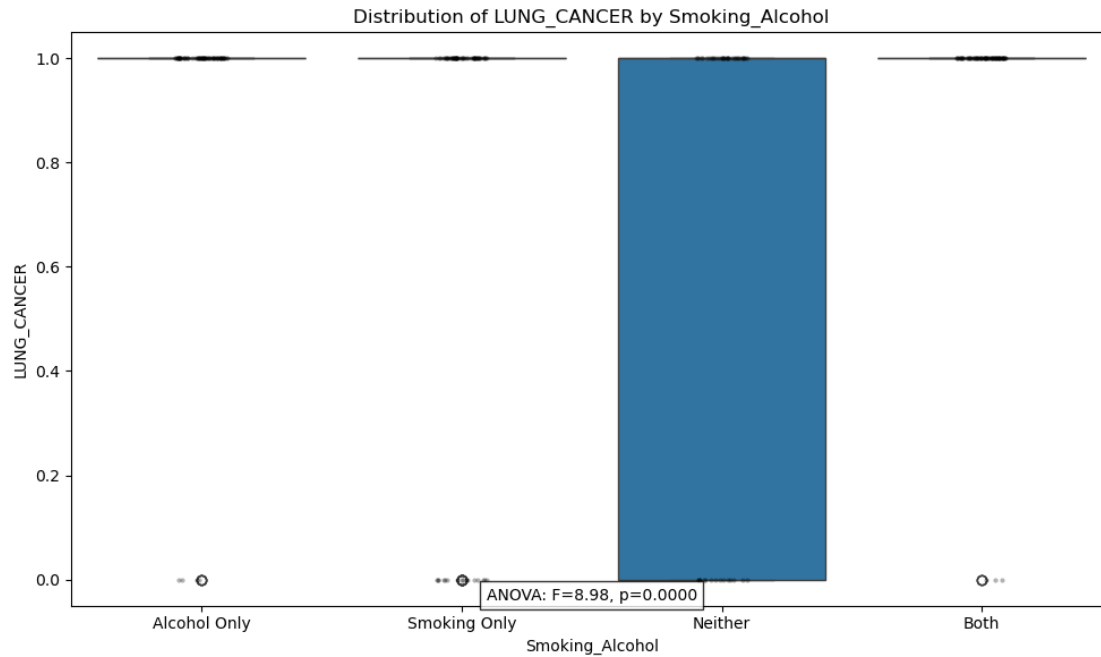
Both vs Neither: Mean diff = -0.2450, $p = 0.0002$

Both vs Smoking Only: Mean diff = -0.1874, $p = 0.0035$

Neither vs Smoking Only: Mean diff = 0.0577, $p = 0.7725$

Box Plots by Group



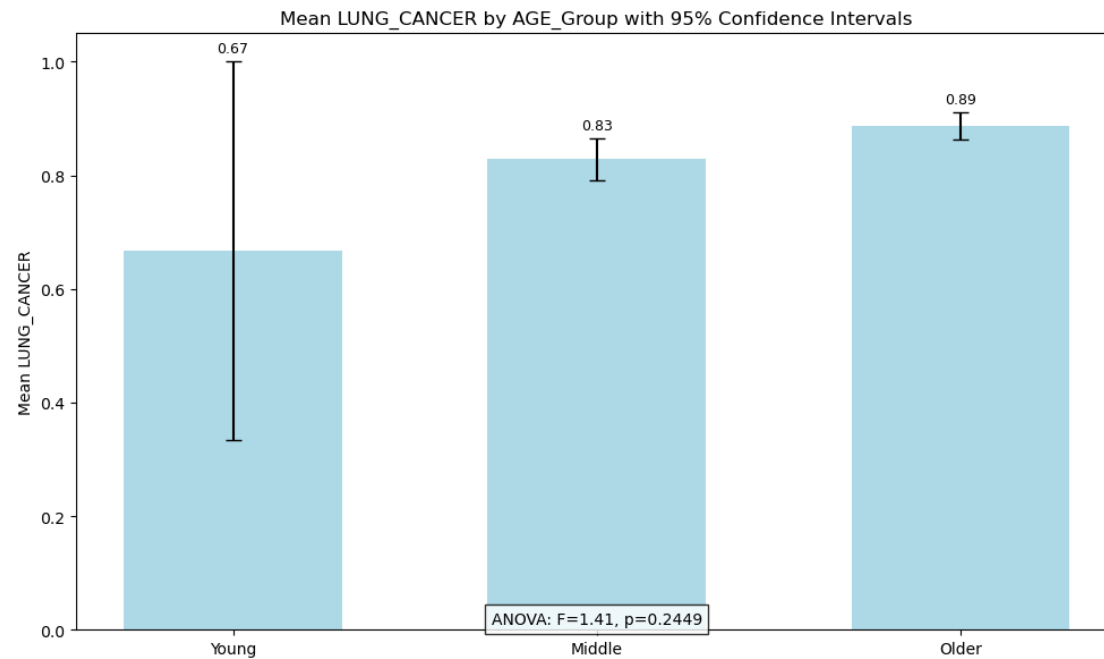


Mean Comparisons

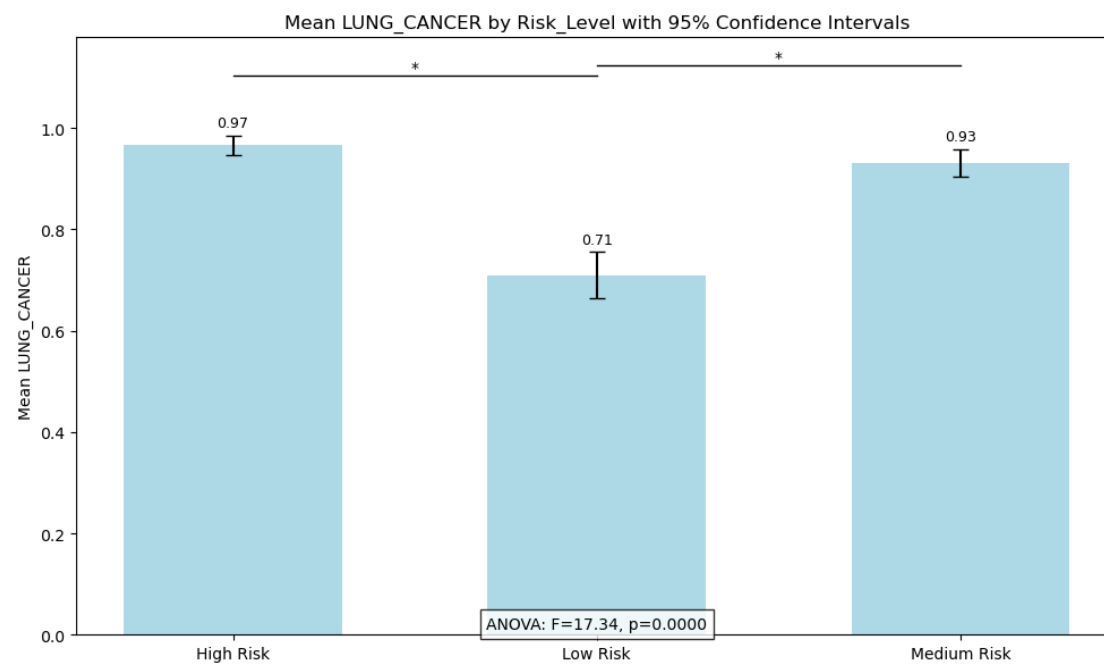
c:\Users\princ\source\repos\school\Reaearch

Methods\preassessment\Isaac\inferential_statistics.py:542: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

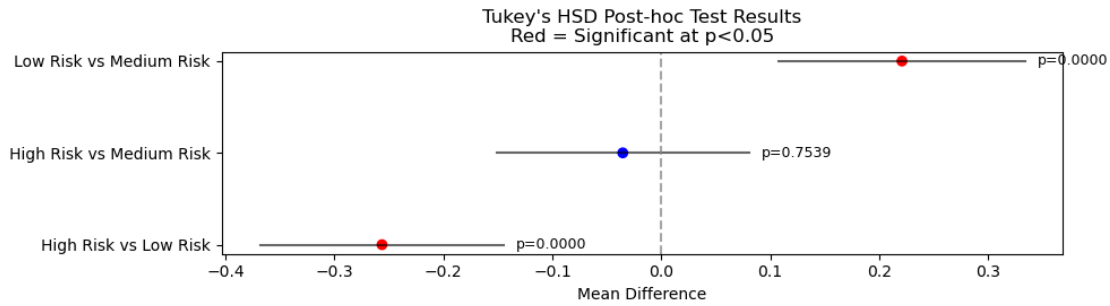
```
group_stats =
anova_data.groupby(actual_group_var)[dependent_var].agg(['mean', 'std',
'count'])
```



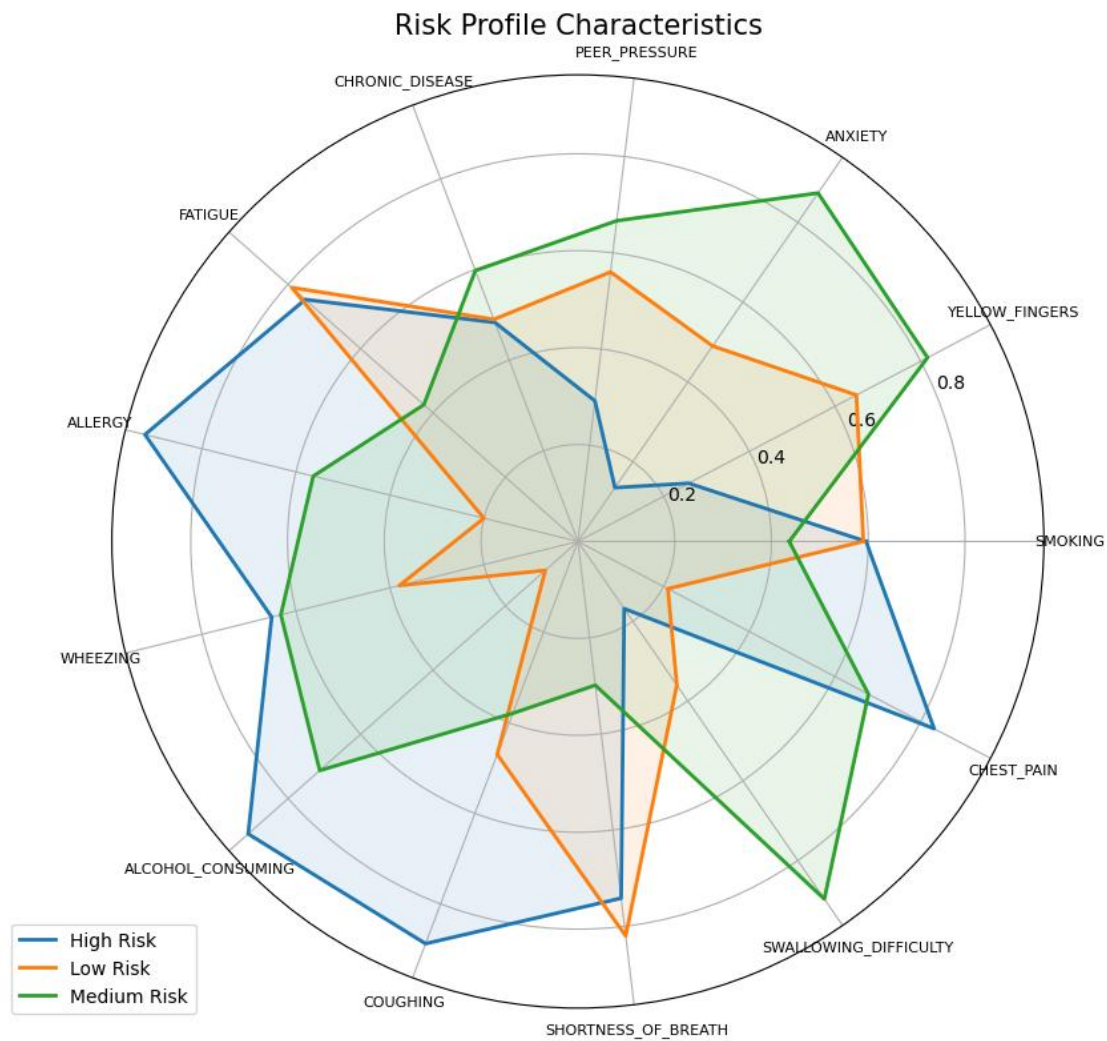
```
posthoc.data[i]: 1  
posthoc.data[i]: 0
```



Post-hoc Test Results



Risk Profile Characteristics



Significant Group Differences Found

Grouping Variable	F-value	p-value	Significant Pairwise Comparisons
Risk_Level	17.34	0.0000	High Risk vs Low Risk, Low Risk vs Medium Risk

| Smoking_Alcohol | 8.98 | 0.0000 | Alcohol Only vs Neither, Alcohol Only vs Smoking Only, Both vs Neither, Both vs Smoking Only |

Bayesian Analysis

Bayesian Logistic Regression

```
print("\n Bayesian Logistic Regression Analysis")
from bayesian_statistics import bayesian_statistics as bayes_stats

trace, bayes_results, bayes_viz =
bayes_stats.bayesian_regression_analysis(data, chains=4, draws=2000,
tune=5000, target_accept=0.95)

print("\n Posterior Distributions")
posterior_fig = bayes_viz['posterior_distributions']()
plt.show()

print("\n Coefficient Estimates with 95% Credible Intervals")
coef_fig = bayes_viz['coefficient_intervals']()
plt.show()

print("\n MCMC Trace Diagnostics")
trace_fig = bayes_viz['trace_diagnostics']()
plt.show()

print("\n Model Comparison (WAIC)")
if 'waic_comparison' in bayes_viz:
    waic_fig = bayes_viz['waic_comparison']()
    plt.show()
else:
    print("No WAIC comparison available.")

print("\n LOO Model Comparison")
if 'loo_comparison' in bayes_viz:
    loo_fig = bayes_viz['loo_comparison']()
    plt.show()
else:
    print("No LOO comparison available.")

print("\n Interaction Effect Sizes")
interact_fig = bayes_viz['interaction_effects']()
plt.show()

print("\n ROC Curve with Credible Interval")
roc_fig = bayes_viz['roc_curve']()
plt.show()

print("\n Confusion Matrix")
cm_fig = bayes_viz['confusion_matrix']()
```

```

plt.show()

print("\n Parameter Estimates with 95% Credible Intervals")
ci_table = bayes_viz['credible_intervals']()

print("\n Main Effects")
print("| Variable | Mean Coefficient | 95% CI | Odds Ratio | OR 95% CI | Significant |")
print("|-----|-----|-----|-----|-----|-----|")
for idx, row in ci_table[ci_table['Type'] == 'Main Effect'].iterrows():
    if idx == 'Intercept':
        continue
    sig_marker = "✓" if row['Significant'] else "X"
    print(f"| {idx} | {row['Mean']:.4f} | {row['95% CI']} | {row['Odds Ratio']:.4f} | {row['OR 95% CI']} | {sig_marker} |")

print("\n Interaction Terms")
print("| Variable | Mean Coefficient | 95% CI | Odds Ratio | OR 95% CI | Significant |")
print("|-----|-----|-----|-----|-----|-----|")
for idx, row in ci_table[ci_table['Type'] == 'Interaction'].iterrows():
    sig_marker = "✓" if row['Significant'] else "X"
    if row['Significant']:
        print(f"| **{idx}** | **{row['Mean']:.4f}** | **{row['95% CI']}** | **{row['Odds Ratio']:.4f}** | **{row['OR 95% CI']}** | **{sig_marker}** |")
    else:
        print(f"| {idx} | {row['Mean']:.4f} | {row['95% CI']} | {row['Odds Ratio']:.4f} | {row['OR 95% CI']} | {sig_marker} |")

print("\n Bayesian Analysis Interpretation")
print("1. **Posterior Distributions**: Show the complete probability distributions for each parameter")
print("2. **Credible Intervals**: Unlike frequentist confidence intervals, credible intervals directly show probability ranges for parameters")
print("3. **Odds Ratios**: Values > 1 indicate increased odds of lung cancer; values < 1 indicate decreased odds")
print("4. **Model Comparison**: WAIC measures predictive accuracy with a penalty for complexity; lower is better")
print("5. **Significant Effects**: Parameters whose 95% credible intervals exclude zero/one (for coefficients/odds ratios)")

```

Bayesian Logistic Regression Analysis

Bayesian Logistic Regression Analysis

Creating Interaction Terms

Fitting Base Bayesian Model (Main Effects Only)

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (4 chains in 4 jobs)

NUTS: [Intercept, AGE_C, GENDER, SMOKING, ALCOHOL_CONSUMING, ANXIETY, PEER_PRESSURE]

```
{"model_id": "7f181533c89c4476a9bad5f1c0d0bf76", "version_major": 2, "version_minor": 0}
```

Sampling 4 chains for 5_000 tune and 2_000 draw iterations (20_000 + 8_000 draws total) took 35 seconds.

```
{"model_id": "26aa39d230fd463fa87aa372408dcd57", "version_major": 2, "version_minor": 0}
```

Fitting Interaction Bayesian Model

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (4 chains in 4 jobs)

NUTS: [Intercept, AGE_C, GENDER, SMOKING, ALCOHOL_CONSUMING, ANXIETY, PEER_PRESSURE, SMOKING_ANXIETY, SMOKING_PEER_PRESSURE, ALCOHOL_ANXIETY, ALCOHOL_PEER_PRESSURE]

```
{"model_id": "6a30d4acdf4f45e08fdff61e77a625b7", "version_major": 2, "version_minor": 0}
```

Sampling 4 chains for 5_000 tune and 2_000 draw iterations (20_000 + 8_000 draws total) took 43 seconds.

```
{"model_id": "791e87d962b2454c8781054a5fe4c96d", "version_major": 2, "version_minor": 0}
```

Fitting Full Bayesian Model (with Age Moderations)

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (4 chains in 4 jobs)

NUTS: [Intercept, AGE_C, GENDER, SMOKING, ALCOHOL_CONSUMING, ANXIETY, PEER_PRESSURE, SMOKING_ANXIETY, SMOKING_PEER_PRESSURE, ALCOHOL_ANXIETY, ALCOHOL_PEER_PRESSURE, AGE_SMOKING, AGE_ALCOHOL, AGE_ANXIETY, AGE_PEER_PRESSURE, GENDER_SMOKING, GENDER_ALCOHOL, GENDER_ANXIETY, GENDER_PEER_PRESSURE]

```
{"model_id":"e4217f618e854db0bffa9af7c456b02","version_major":2,"version_minor":0}
```

Sampling 4 chains for 5_000 tune and 2_000 draw iterations (20_000 + 8_000 draws total) took 57 seconds.

```
{"model_id":"b8c7212829314a8392b140036494290a","version_major":2,"version_minor":0}
```

Calculating LOO-CV

LOO Comparison						
	rank	elpd_loo	p_loo	elpd_diff	weight	se
\						
interaction	0	182.960204	7.079819	0.000000	9.198950e-01	20.408474
full	1	183.767482	9.585055	0.807278	8.010504e-02	20.340710
base	2	185.190803	6.024573	2.230599	6.661338e-16	21.038194

	dse	warning	scale
interaction	0.000000	False	deviance
full	1.966416	False	deviance
base	1.765703	False	deviance

Selected model based on LOO: interaction

Selecting final model based on LOO rank 0: interaction

Model Comparison Summary
Base Model WAIC: -92.57
Interaction Model WAIC: -91.45
Full Model WAIC: -91.78

LOO Comparison Table:

	rank	elpd_loo	p_loo	elpd_diff	weight	se
--	------	----------	-------	-----------	--------	----

\						
interaction	0	182.960204	7.079819	0.000000	9.198950e-01	20.408474
full	1	183.767482	9.585055	0.807278	8.010504e-02	20.340710
base	2	185.190803	6.024573	2.230599	6.661338e-16	21.038194

	dse	warning	scale
interaction	0.000000	False	deviance
full	1.966416	False	deviance
base	1.765703	False	deviance

Approximate Bayes Factors
Interaction vs Base: 13895.77
Full vs Interaction: 2004968610.72

Proceeding with selected model: interaction

Extracted Posterior Samples

Using 500 samples for prediction

Calculated Predicted Probabilities

Model Evaluation
AUC: 0.8288

Classification Report					
	precision	recall	f1-score	support	
0	0.50	0.08	0.14	38	
1	0.87	0.99	0.93	238	
accuracy			0.86	276	
macro avg	0.69	0.53	0.53	276	
weighted avg	0.82	0.86	0.82	276	

Posterior Distributions
Intervals are:

```
<xarray.Dataset>
Dimensions:      (hdi: 2)
Coordinates:
  * hdi          (hdi) <U6 'lower' 'higher'
Data variables:
  Intercept      (hdi) float64 -0.3798 1.018
Intercept: [-0.380, 1.018]
Intervals are:
```

```
<xarray.Dataset>
```

```
Dimensions: (hdi: 2)
Coordinates:
  * hdi      (hdi) <U6 'lower' 'higher'
Data variables:
  SMOKING    (hdi) float64 -0.864 0.6741
SMOKING: [-0.864, 0.674]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  ALCOHOL_CONSUMING (hdi) float64 1.334 3.046
ALCOHOL_CONSUMING: [1.334, 3.046]
Intervals are:
```

```
<xarray.Dataset>
Dimensions: (hdi: 2)
Coordinates:
  * hdi      (hdi) <U6 'lower' 'higher'
Data variables:
  ANXIETY    (hdi) float64 -0.1432 1.472
ANXIETY: [-0.143, 1.472]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  PEER_PRESSURE (hdi) float64 0.2702 1.912
PEER_PRESSURE: [0.270, 1.912]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  ALCOHOL_CONSUMING (hdi) float64 1.334 3.046
ALCOHOL_CONSUMING: [1.334, 3.046]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
```

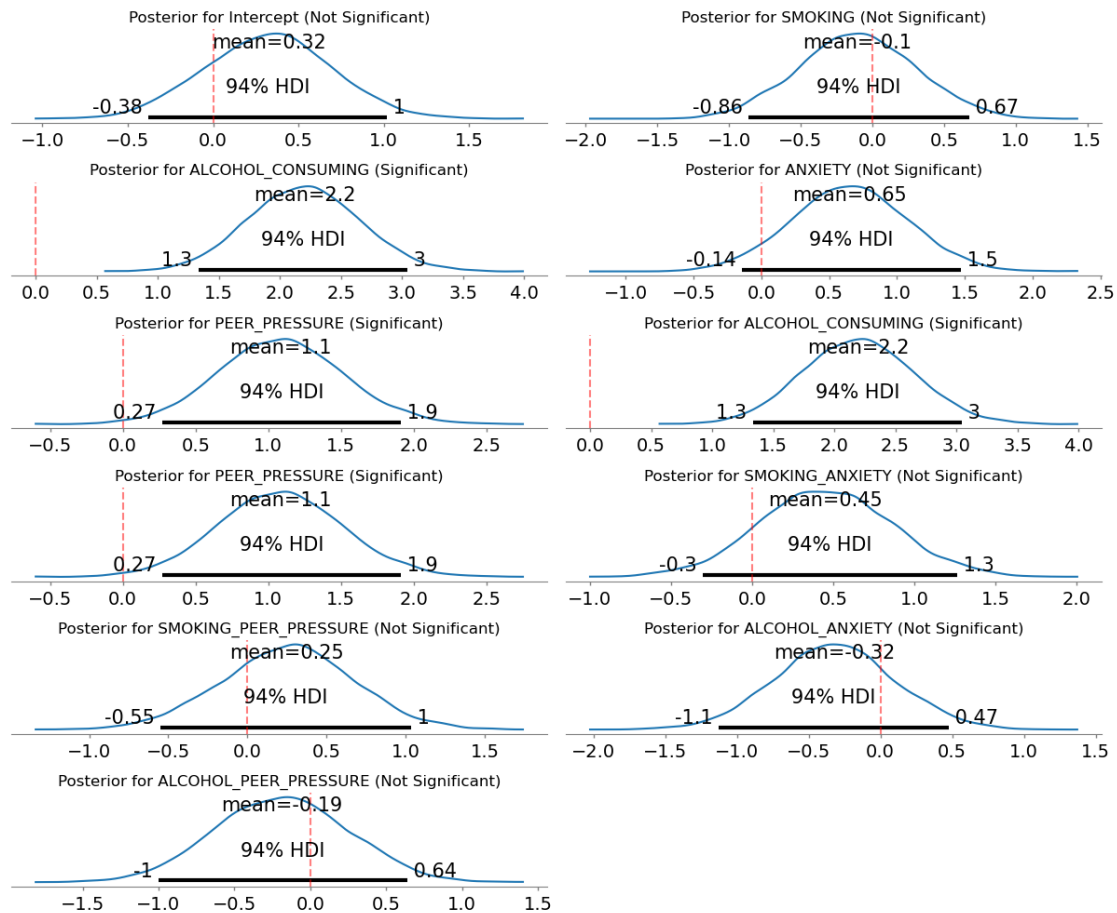
```
PEER_PRESSURE (hdi) float64 0.2702 1.912
PEER_PRESSURE: [0.270, 1.912]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  SMOKING_ANXIETY    (hdi) float64 -0.3027 1.262
SMOKING_ANXIETY: [-0.303, 1.262]
Intervals are:
```

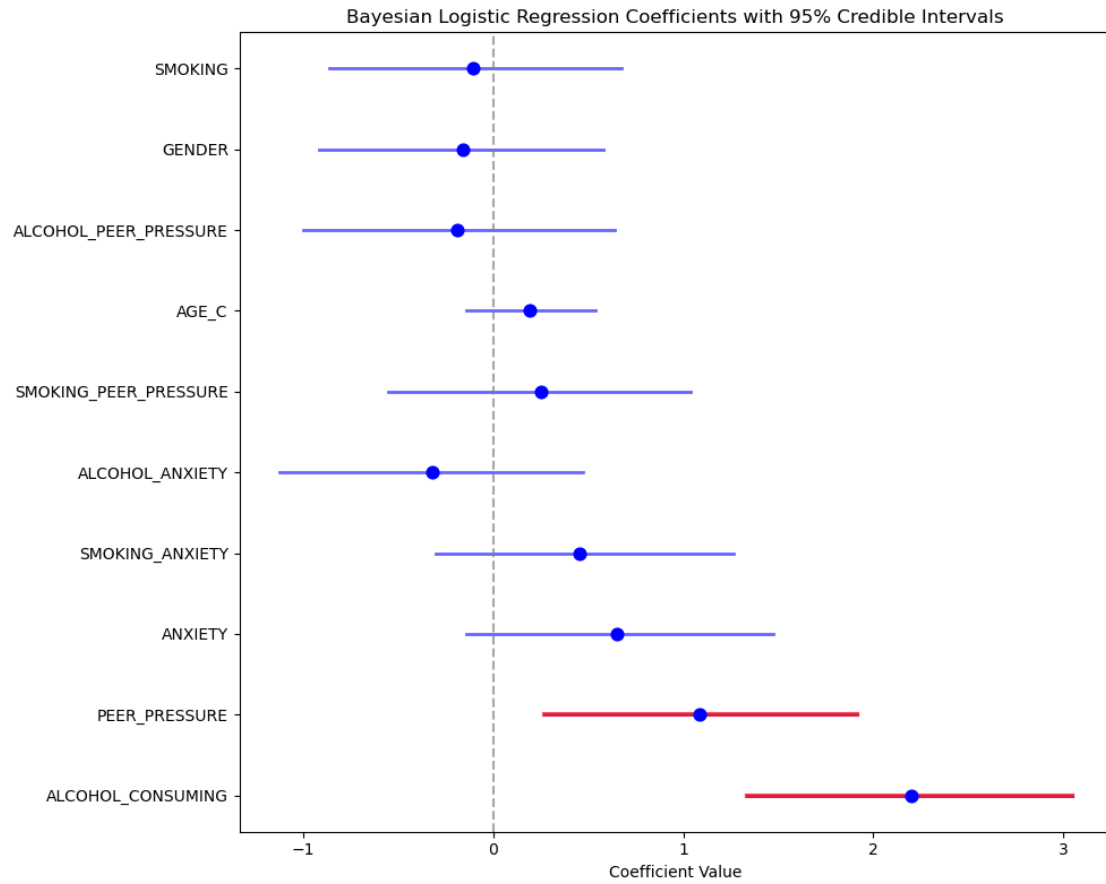
```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  SMOKING_PEER_PRESSURE (hdi) float64 -0.5496 1.036
SMOKING_PEER_PRESSURE: [-0.550, 1.036]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  ALCOHOL_ANXIETY      (hdi) float64 -1.126 0.4742
ALCOHOL_ANXIETY: [-1.126, 0.474]
Intervals are:
```

```
<xarray.Dataset>
Dimensions:          (hdi: 2)
Coordinates:
  * hdi              (hdi) <U6 'lower' 'higher'
Data variables:
  ALCOHOL_PEER_PRESSURE (hdi) float64 -1.0 0.6387
ALCOHOL_PEER_PRESSURE: [-1.000, 0.639]
```

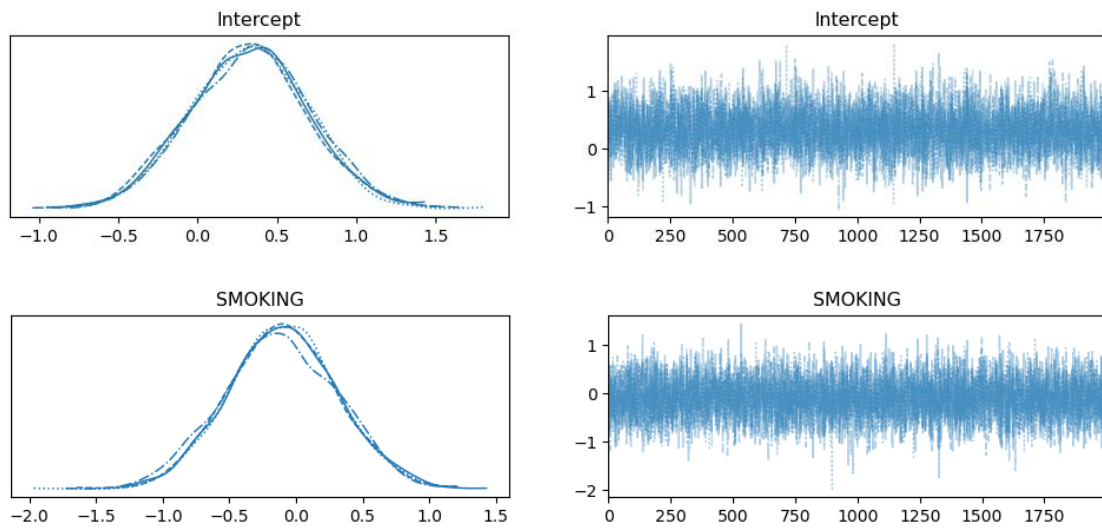



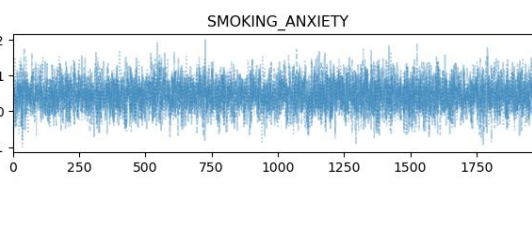
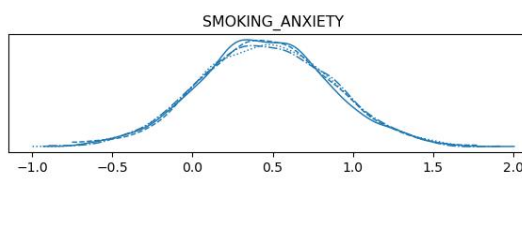
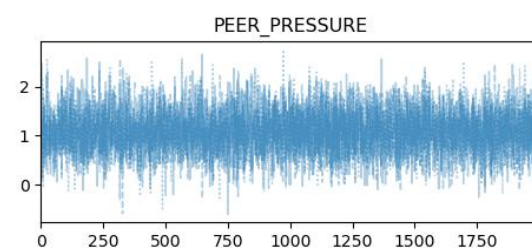
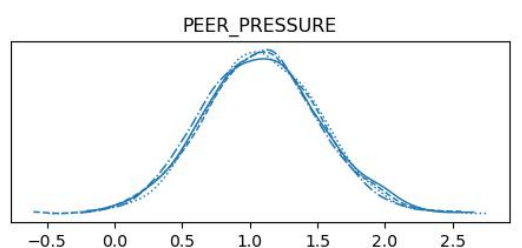
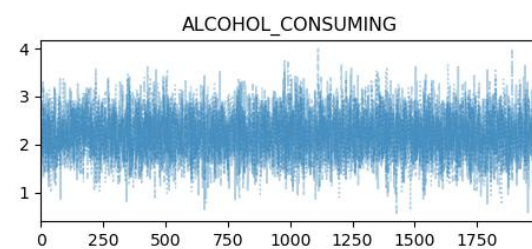
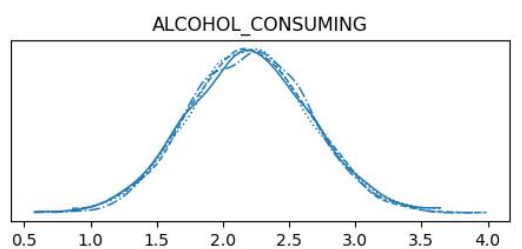
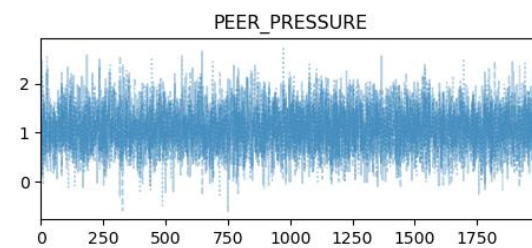
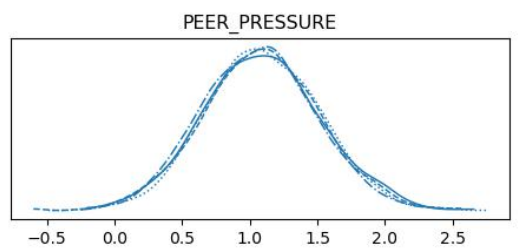
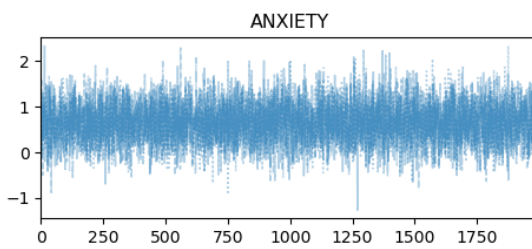
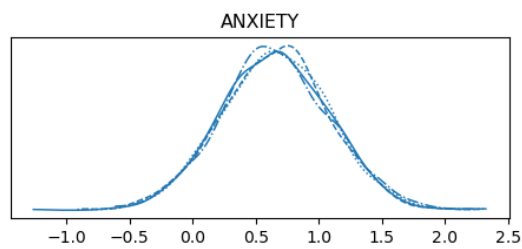
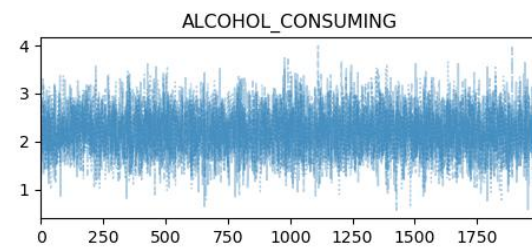
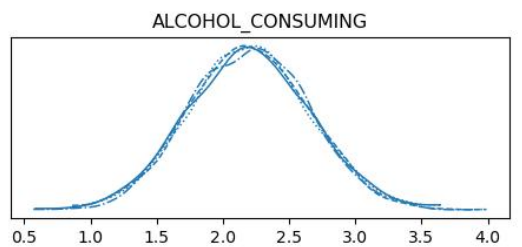
Coefficient Estimates with 95% Credible Intervals



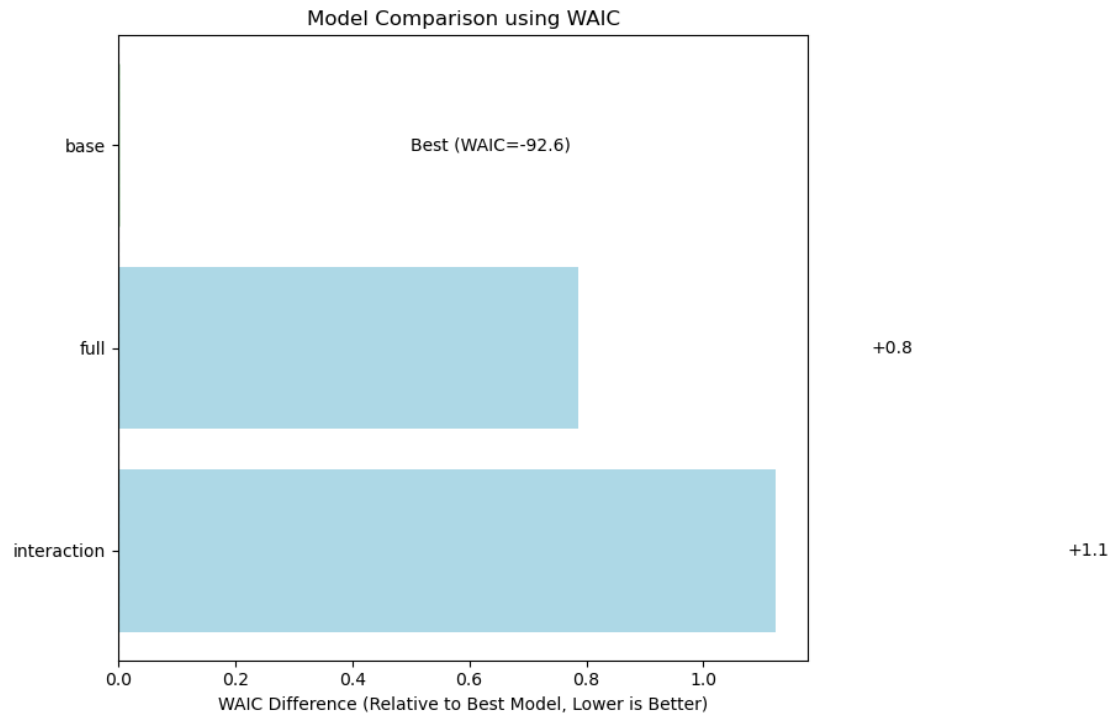
MCMC Trace Diagnostics

<Figure size 1200x1000 with 0 Axes>

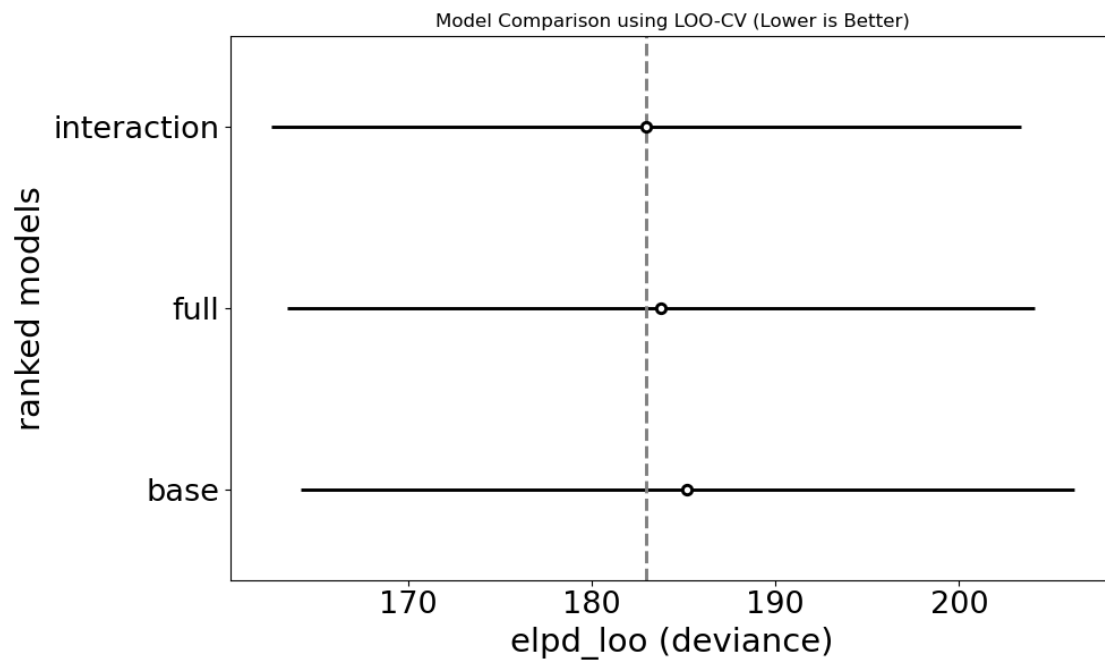




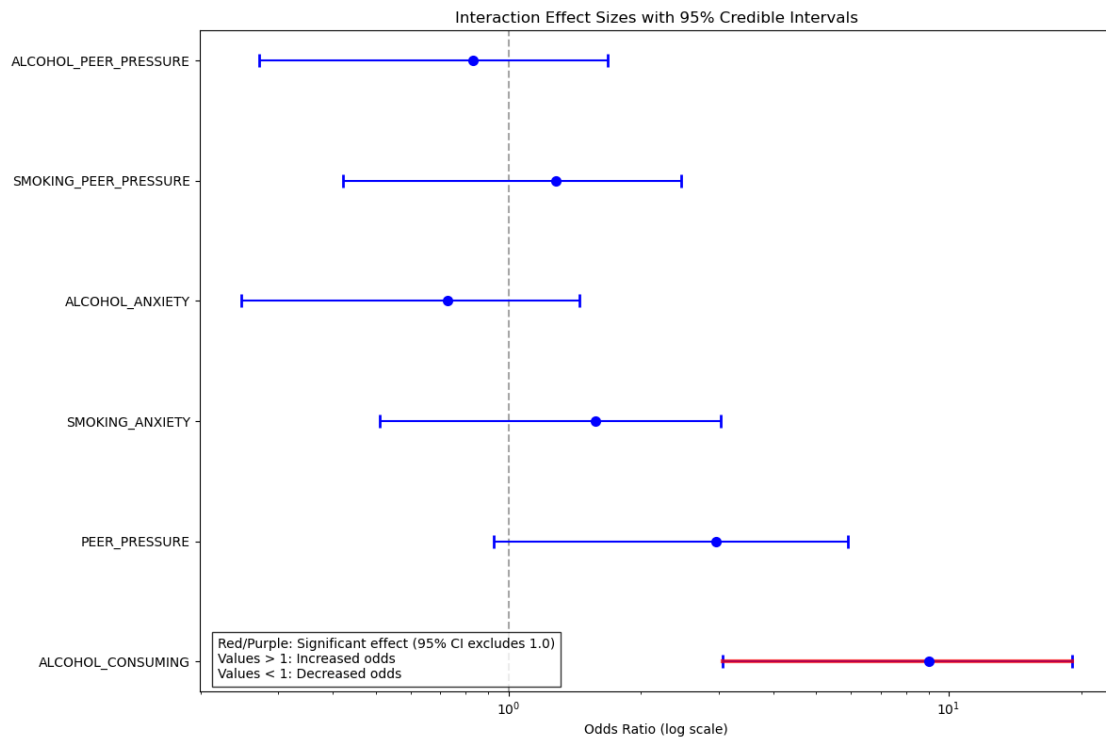
Model Comparison (WAIC)



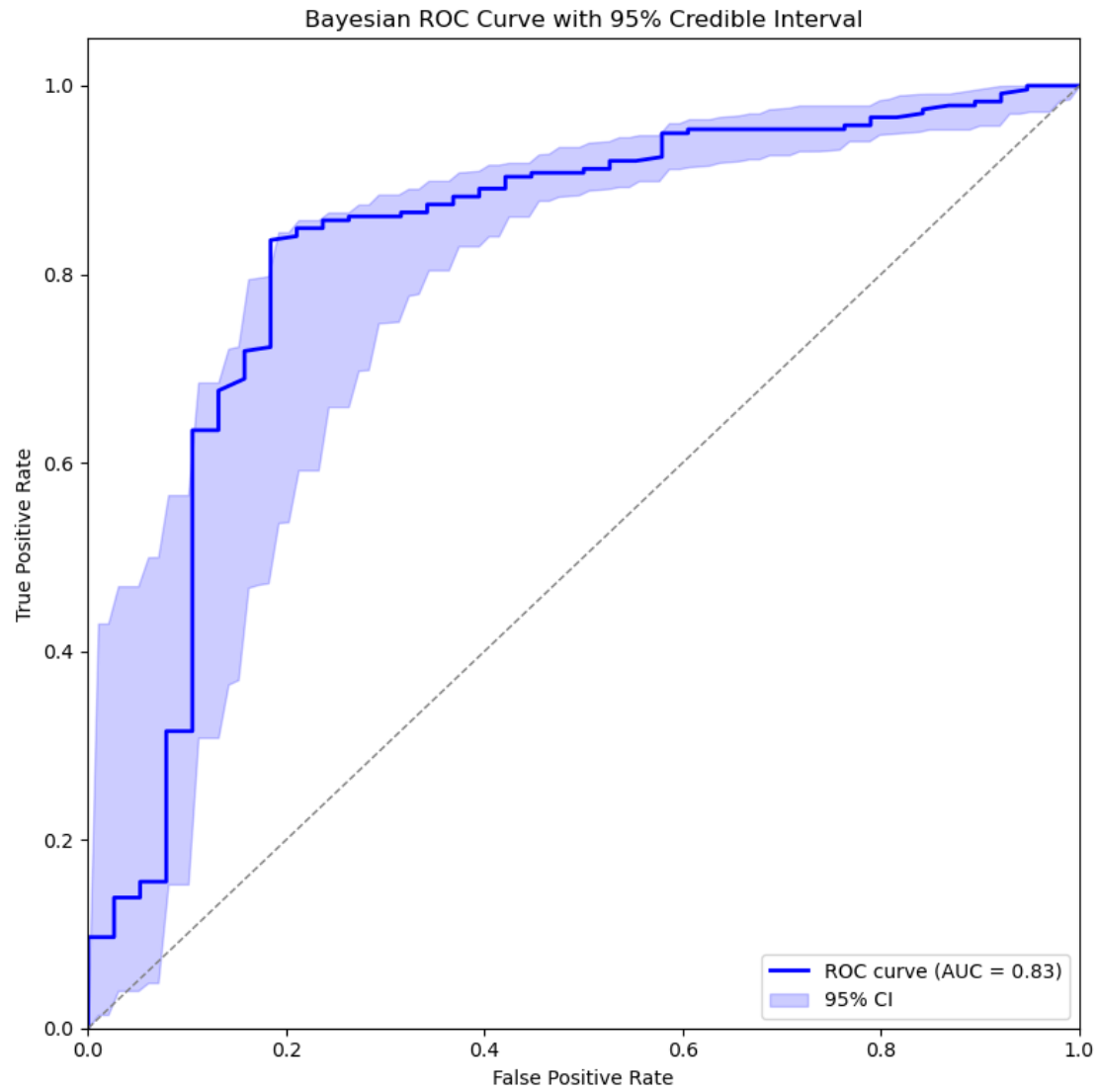
LOO Model Comparison



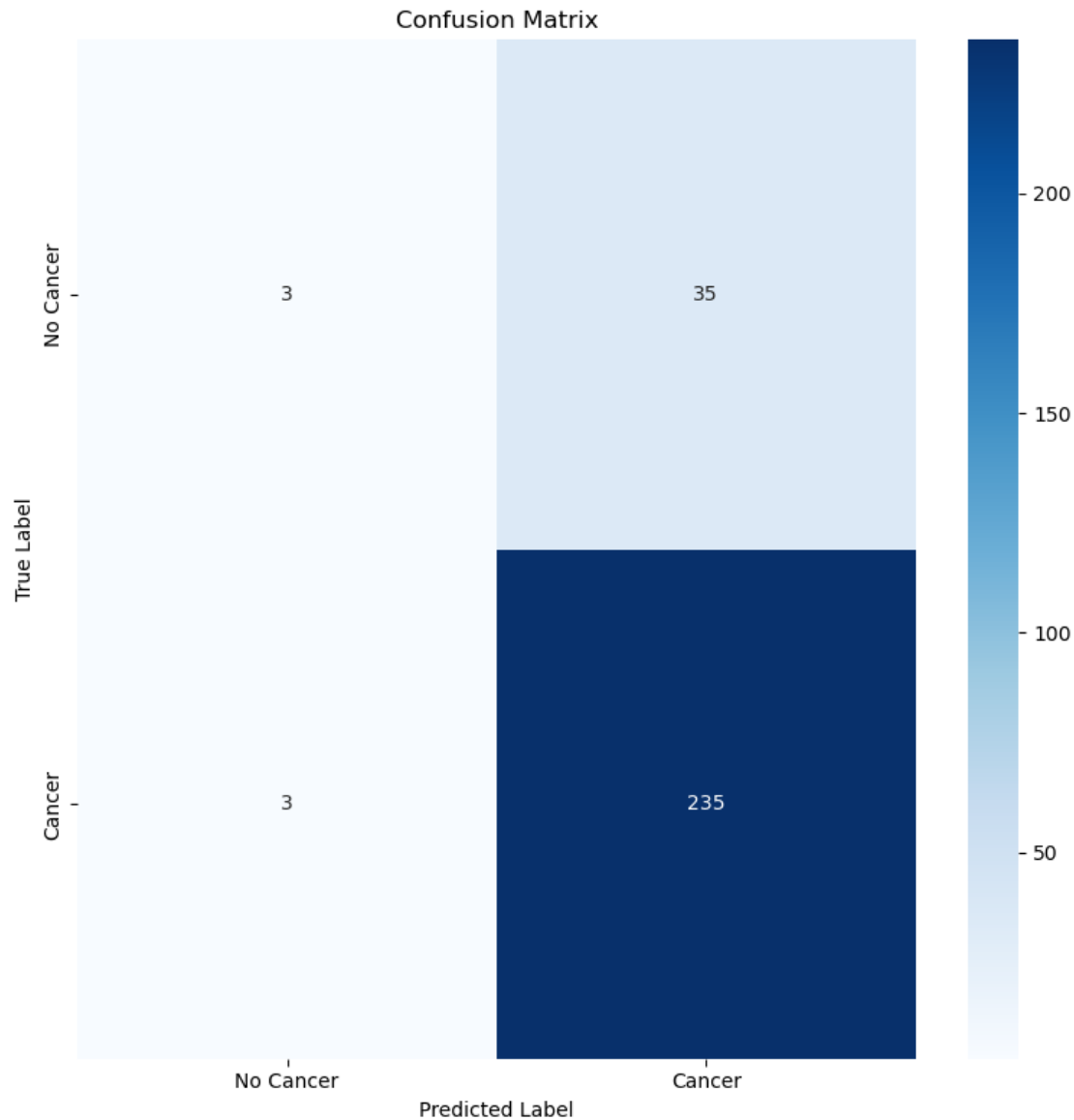
Interaction Effect Sizes



ROC Curve with Credible Interval



Confusion Matrix



Parameter Estimates with 95% Credible Intervals

Main Effects

Variable	Mean Coefficient	95% CI	Odds Ratio	OR 95% CI	Significant
ANXIETY	0.6531	(-0.1923, 1.4972)	2.1105	(0.8250, 4.4693)	X
AGE_C	0.1908	(-0.1599, 0.5504)	1.2306	(0.8522, 1.7339)	X
SMOKING	-0.1038	(-0.9023, 0.7044)	0.9810	(0.4056, 2.0226)	X
GENDER	-0.1606	(-0.9250, 0.6269)	0.9229	(0.3965, 1.8718)	X

Interaction Terms

Variable	Mean Coefficient	95% CI	Odds Ratio	OR 95% CI	Significant
ALCOHOL_CONSUMING	2.2004	(1.3175, 3.1044)	10.0398	(3.7341, 22.2956)	✓
PEER_PRESSURE	1.0836	(0.2442, 1.9538)	3.2513	(1.2767, 7.0557)	✓
SMOKING_ANXIETY	0.4537	(-0.3670, 1.2727)	1.7142	(0.6928, 3.5705)	X
SMOKING_PEER_PRESSURE	0.2499	(-0.5939, 1.0779)	1.4024	(0.5522, 2.9384)	X
ALCOHOL_PEER_PRESSURE	-0.1869	(-1.0422, 0.6674)	0.9147	(0.3527, 1.9491)	X
ALCOHOL_ANXIETY	-0.3190	(-1.1583, 0.5205)	0.7960	(0.3140, 1.6829)	X

Bayesian Analysis Interpretation

- Posterior Distributions**: Show the complete probability distributions for each parameter
- Credible Intervals**: Unlike frequentist confidence intervals, credible intervals directly show probability ranges for parameters
- Odds Ratios**: Values > 1 indicate increased odds of lung cancer; values < 1 indicate decreased odds
- Model Comparison**: WAIC measures predictive accuracy with a penalty for complexity; lower is better
- Significant Effects**: Parameters whose 95% credible intervals exclude zero/one (for coefficients/odds ratios)